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LAB MANUAL

UIT2702 Cloud and Distributed Computing

Student Name: Vaibhav SP

Register Number: 3122235002146

Batch / Section: IT-C

Faculty / Faculty In-charge: Dr. Sabireen H

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Exercise 1: Inter-Process Communication in Distributed Computing

AIM:

To implement and understand inter-process communication using sockets in a distributed client-server setup.

PROCEDURE:

Task 1 - Setup Basic Client-Server Communication

- Create a simple socket connection between the client and server.
- On the server side, create a TCP socket and bind it to host 127.0.0.1 and port 5000.
- Make the server listen for a client connection and accept the connection.
- On the client side, create a socket and connect to the server using the same host and port.
- Enter numbers separated by spaces on the client and send the message to the server.
- Receive the message on the server, decode it, split the numbers, and convert them to integers.
- Calculate the sum of the received numbers on the server.
- Send the calculated sum back to the client and display the server response.

Task 2 - Enhanced Communication with Object Serialization

- Define a custom DataObject containing the name and values attributes.
- Serialize the DataObject using Python pickle before sending it from the client.
- Send the serialized object from the client to the server.
- Deserialize the received object on the server using pickle.
- Read the values from the DataObject and calculate their sum on the server.
- Send a response containing the name and sum of values back to the client.
- Display the received response on the client.

PROGRAM / SOURCE CODE:

server.py

```
import socket
HOST = '127.0.0.1'
PORT = 5000
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)
print("Server is waiting for connection...")
conn, addr = server.accept()
print("Connected by:", addr)
data = conn.recv(1024).decode()
numbers = list(map(int, data.split()))
result = sum(numbers)
message = f"Sum = {result}"
conn.send(message.encode())
conn.close()
server.close()
```

client.py

```
import socket
HOST = '127.0.0.1'
PORT = 5000
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))
msg = input("Enter numbers separated by space: ")
client.send(msg.encode())
response = client.recv(1024).decode()
print("Server Response:", response)
client.close()
```

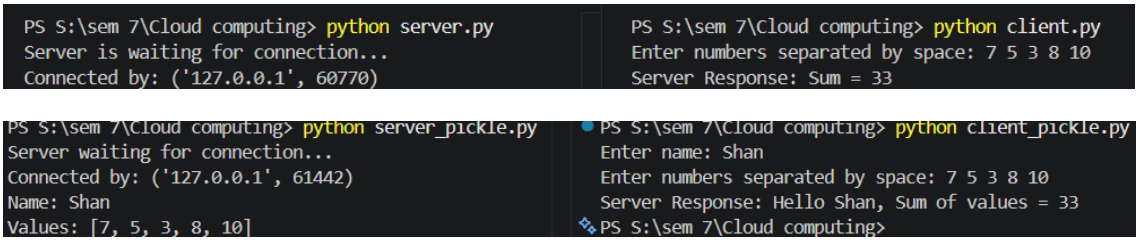
server_pickle.py

```
import socket
import pickle
class DataObject:
    def __init__(self, name, values):
        self.name = name
        self.values = values
HOST = '127.0.0.1'
PORT = 5001
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)
print("Server waiting for connection...")
conn, addr = server.accept()
print("Connected by:", addr)
data = conn.recv(4096)
obj = pickle.loads(data)
print("Name:", obj.name)
print("Values:", obj.values)
total = sum(obj.values)
response = f"Hello {obj.name}, Sum of values = {total}"
conn.send(pickle.dumps(response))
conn.close()
server.close()
```

client_pickle.py

```
import socket
import pickle
class DataObject:
    def __init__(self, name, values):
        self.name = name
        self.values = values
HOST = '127.0.0.1'
PORT = 5001
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))
name = input("Enter name: ")
values = list(map(int, input("Enter numbers separated by space: ").split()))
obj = DataObject(name, values)
client.send(pickle.dumps(obj))
response = pickle.loads(client.recv(4096))
print("Server Response:", response)
client.close()
```

SCREENSHOTS:



OUTPUT:

Task 1:

Server is waiting for connection...

Connected by: (127.0.0.1)

Server Response: Sum = 33

Task 2:

Name: Shan

Values: [7, 5, 3, 8, 10]

Server Response: Hello Shan, Sum of values = 33

RESULT:

Inter-process communication using sockets in a distributed client-server setup is implemented successfully.

Exercise 2: Remote Method Invocation (RMI)

AIM:

To implement Remote Method Invocation (RMI) to facilitate remote execution of methods in a distributed computing setup.

PROCEDURE:

Task 1 - Define and Expose the Remote Service

- Create the Calculator class on the server.
- Define methods for addition and multiplication in the Calculator class.
- Use Pyro4 to expose the Calculator methods so that they can be invoked remotely.

Task 2 - Set Up the RMI Server

- Create a Pyro4.Daemon on the server.
- Register the Calculator service with the daemon.
- Obtain the unique URI generated for the registered Calculator service.
- Start the daemon so that it listens for incoming client requests.

Task 3 - Invoke the Remote Methods from the Client

- Run the client and provide the URI of the remote Calculator service.
- Create a Pyro4.Proxy using the supplied URI.
- Invoke the remote add_numbers and multiply methods through the proxy.
- Display the addition and multiplication results on the client side.

PROGRAM / SOURCE CODE:

rmi_server.py

```
import Pyro4
@Pyro4.expose
class Calculator:
    def add(self, a, b):
        return a + b
    def multiply(self, a, b):
        return a * b
daemon = Pyro4.Daemon()
uri = daemon.register(Calculator)
print("Server is running...")
print("URI:", uri)
daemon.requestLoop()
```

rmi_client.py

```
import Pyro4
uri = input("Enter the server URI: ")
calculator = Pyro4.Proxy(uri)
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
addition = calculator.add(a, b)
multiplication = calculator.multiply(a, b)
print("Addition =", addition)
print("Multiplication =", multiplication)
```

SCREENSHOTS:

```
PS S:\sem 7\python rmi_server.py
Server is running...
URI: PYRO:obj_05e1f5f02f104eb6b8634bbf75c635ab@localhost:56618
```

```
PS S:\sem 7\python rmi_client.py
Enter the server URI: PYRO:obj_05e1f5f02f104eb6b8634bbf75c635ab@localhost:56618
Enter first number: 7
Enter second number: 5
Addition = 12
Multiplication = 35
```

OUTPUT:

Server is running...

URI: PYRO:0bj_05e1f5f02f104eb6b8634bbf75c635ab@localhost:56618

Addition = 12

Multiplication = 35

RESULT:

Remote Method Invocation (RMI) to facilitate remote execution of methods in a distributed computing setup is implemented successfully.

Exercise 3: Message passing and distributed mutual exclusion

AIM:

To implement message passing and distributed mutual exclusion using sockets.

PROCEDURE:

Task 1 - Message Passing System

- Create a server that listens for socket connections from multiple clients.
- Accept client connections and maintain the connected clients for communication.
- Allow each client to send messages to the server.
- Forward the messages so that connected clients can send and receive messages.
- Verify communication by sending messages between the connected clients and observing the responses.

Task 2 - Distributed Mutual Exclusion

- Allow a process to request access to the critical section by sending a message.
- Process the request according to the selected mutual exclusion algorithm.
- Grant access to the critical section only when the algorithm permits the requesting process to enter.
- Ensure that only one process enters the critical section at a time.
- After the critical section is completed, release access using message passing.
- Implement the Ricart-Agrawala algorithm for requesting, granting, and releasing access to the critical section.
- Implement the Token Ring algorithm so that the process holding the token can enter the critical section and then pass the token to the next process.
- Implement the Centralized Coordinator algorithm so that requests are sent to the coordinator, access is granted to one process, and the process sends a release after leaving the critical section.

PROGRAM / SOURCE CODE:

server.py

```
import socket
import threading
clients = []
def broadcast(message, sender=None):
    for client in clients[:]:
        if client != sender:
            try:
                client.send(message.encode())
            except:
                clients.remove(client)
def handle_client(conn, addr):
    name = conn.recv(1024).decode()
    print(name, "joined")
    broadcast(name + " joined the chat", conn)
    while True:
        try:
            msg = conn.recv(1024).decode()
```

```
            if not msg:
                break
            print(name + ":", msg)
            broadcast(name + ": " + msg, conn)
        except:
            break
    if conn in clients:
        clients.remove(conn)
    conn.close()
    print(name, "left")
    broadcast(name + " left the chat")
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("localhost", 6000))
server.listen()
server.settimeout(1)      # Check every 1 second
print("Server started... Press Ctrl+C to stop.")
try:
    while True:
        try:
            conn, addr = server.accept()
            clients.append(conn)
            threading.Thread(target=handle_client, args=(conn, addr), daemon=True).start()
        except socket.timeout:
            continue
except KeyboardInterrupt:
    print("\nStopping server...")
finally:
    for client in clients:
        client.close()
    server.close()
print("Server stopped.")
```

client.py

```
import socket
import threading
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("localhost", 6000))
name = input("Enter your name: ")
client.send(name.encode())
def receive():
    while True:
        try:
            print(client.recv(1024).decode())
        except:
            break
threading.Thread(target=receive, daemon=True).start()
print("Type messages (type 'exit' to quit)")
while True:
    msg = input()
    if msg == "exit":
        break
    client.send(msg.encode())
client.close()
```

ricart_agrawala.py

```
import socket
import threading
import time
import sys
ME = int(sys.argv[1])
N = 3
PORT = 7000 + ME
replies = 0
requesting = False
deferred = []
```

```
def send(pid, msg):
    try:
        s = socket.socket()
        s.connect(("localhost", 7000 + pid))
        s.send(msg.encode())
        s.close()
    except:
        pass
def handle(msg):
    global replies
    parts = msg.split()
    if parts[0] == "REQUEST":
        pid = int(parts[1])
        print("REQUEST from P%d" % pid)
        if requesting:
            deferred.append(pid)
        else:
            send(pid, "REPLY %d" % ME)
    elif parts[0] == "REPLY":
        replies += 1
        print("REPLY from P%d (%d/%d)" % (int(parts[1]), replies, N - 1))
def listener():
    server = socket.socket()
    server.bind(("localhost", PORT))
    server.listen(5)
    server.settimeout(1)
    try:
        while True:
            try:
                conn, addr = server.accept()
                msg = conn.recv(1024).decode()
                conn.close()
                handle(msg)
            except socket.timeout:
                pass
    except KeyboardInterrupt:
        pass
    finally:
        server.close()
threading.Thread(target=listener, daemon=True).start()
print("Process", ME, "started")
try:
    while True:
        input("Press Enter to request Critical Section...")
        requesting = True
        replies = 0
        for i in range(1, N + 1):
            if i != ME:
                send(i, "REQUEST %d" % ME)
        while replies < N - 1:
            time.sleep(0.1)
        print(">>> P%d ENTERED Critical Section" % ME)
        time.sleep(5)
        print("<<< P%d LEFT Critical Section" % ME)
        requesting = False
        for pid in deferred:
            send(pid, "REPLY %d" % ME)
        deferred.clear()
except KeyboardInterrupt:
    print("\nProcess stopped.")
```

token_ring.py

```
import socket
import threading
import sys
```

```
import time
ME = int(sys.argv[1])
N = 3
NEXT = ME % N + 1
PORT = 8000 + ME
want = False
def send_token():
    try:
        s = socket.socket()
        s.connect(("localhost", 8000 + NEXT))
        s.send(b"TOKEN")
        s.close()
    except:
        pass
def use_token():
    global want
    print("P%d received TOKEN" % ME)
    if want:
        print("P%d ENTERED Critical Section" % ME)
        time.sleep(3)
        print("P%d LEFT Critical Section" % ME)
        want = False
    print("P%d passed TOKEN to P%d" % (ME, NEXT))
    time.sleep(1)
    send_token()
def request_cs():
    global want
    while True:
        try:
            input("Press Enter to request Critical Section...")
            want = True
        except KeyboardInterrupt:
            break
threading.Thread(target=request_cs, daemon=True).start()
server = socket.socket()
server.bind(("localhost", PORT))
server.listen(5)
server.settimeout(1)
print("Process", ME, "started")
# Process 1 starts with the token
if ME == 1:
    time.sleep(2)
    use_token()
try:
    while True:
        try:
            conn, addr = server.accept()
            msg = conn.recv(1024).decode()
            conn.close()
            if msg == "TOKEN":
                use_token()
        except socket.timeout:
            pass
except KeyboardInterrupt:
    print("\nProcess stopped.")
finally:
    server.close()
```

coordinator.py

```
import socket
import threading
clients = {}
queue = []
busy = False
def grant(pid):
```

```
    clients[pid].send(b"GRANT")
    print("GRANT -> P%d" % pid)
def handle(conn):
    global busy
    while True:
        try:
            msg = conn.recv(1024).decode()
            if not msg:
                break
            cmd, pid = msg.split()
            pid = int(pid)
            clients[pid] = conn
            if cmd == "REQUEST":
                print("REQUEST from P%d" % pid)
                if busy:
                    queue.append(pid)
                else:
                    busy = True
                    grant(pid)
            elif cmd == "RELEASE":
                print("RELEASE from P%d" % pid)
                if queue:
                    grant(queue.pop(0))
                else:
                    busy = False
        except:
            break
    conn.close()
server = socket.socket()
server.bind(("localhost", 9000))
server.listen(5)
server.settimeout(1)
print("Coordinator started...")
try:
    while True:
        try:
            conn, addr = server.accept()
            print("Process connected:", addr)
            threading.Thread(target=handle, args=(conn,), daemon=True).start()
        except socket.timeout:
            pass
except KeyboardInterrupt:
    print("\nCoordinator stopped.")
finally:
    for c in clients.values():
        c.close()
    server.close()
```

process.py

```
import socket
import sys
import time
ME = int(sys.argv[1])
client = socket.socket()
client.connect(("localhost", 9000))
print("Process", ME, "connected")
try:
    while True:
        input("Press Enter to request Critical Section...")
        client.send(f"REQUEST {ME}".encode())
        print("REQUEST sent")
        reply = client.recv(1024).decode()
        if reply == "GRANT":
            print(">>> P%d ENTERED Critical Section" % ME)
            time.sleep(5)
```

```
            print("<<< P%d LEFT Critical Section" % ME)
            client.send(f"RELEASE {ME}".encode())
    except KeyboardInterrupt:
        print("\nProcess stopped.")
    finally:
        client.close()
```

SCREENSHOTS:



























OUTPUT:

Task 1:

John: Hello

Jane: Hi there

John left the chat

Task 2 - Ricart-Agrawala Algorithm:

REQUEST sent

>>> P1 ENTERED Critical Section

<<< P1 LEFT Critical Section

Task 2 - Token Ring Algorithm:

P1 passed TOKEN to P2

P2 received TOKEN

P2 passed TOKEN to P3

Task 2 - Centralized Coordinator Algorithm:

GRANT -> P2

RELEASE from P2

REQUEST from P3

GRANT -> P3

RELEASE from P3

Coordinator stopped.

RESULT:

Message passing and distributed mutual exclusion is implemented successfully using sockets.

Exercise 4: Message Passing and Distributed Mutual Exclusion

AIM:

To implement logical clocks and vector clocks to manage event ordering and causality in distributed systems.

PROCEDURE:

Task 1 - Logical Clocks (Lamport's Logical Clock)

- Initialize each process with a logical clock.
- For each event or message sent, increment the clock and send the timestamp.
- On message receipt, update the clock based on the received timestamp, ensuring ordered events.

Task 2 - Vector Clocks

- Set up each process with a vector representing the state of all processes.
- For each event or message sent, update the process's vector and transmit it.
- Upon receipt, merge the vectors to reflect concurrent events and causality relationships.

PROGRAM / SOURCE CODE:

lamport_clock.py

```
class Process:
    def __init__(self, name):
        self.name = name
        self.clock = 0
    def local_event(self):
        self.clock += 1
        print(f"{self.name}: Local event -> Clock = {self.clock}")
    def send_message(self):
        self.clock += 1
        timestamp = self.clock
        print(f"{self.name}: Sending message -> Timestamp = {timestamp}")
        return timestamp
    def receive_message(self, timestamp):
        self.clock = max(self.clock, timestamp) + 1
        print(
            f"{self.name}: Received message "
            f"(Timestamp = {timestamp}) -> Clock = {self.clock}"
        )
# Create two processes
P1 = Process("P1")
P2 = Process("P2")
# Events in P1
P1.local_event()
timestamp = P1.send_message()
# P2 receives message
P2.receive_message(timestamp)
# Another local event in P2
P2.local_event()
# P2 sends message back
timestamp = P2.send_message()
# P1 receives message
P1.receive_message(timestamp)
```

vector_clock.py

```
class VectorProcess:
    def __init__(self, name, process_id, total_processes):
        self.name = name
        self.process_id = process_id
        self.vector = [0] * total_processes
    def local_event(self):
        self.vector[self.process_id] += 1
        print(
            f"{self.name}: Local event -> "
            f"Vector = {self.vector}"
        )
    def send_message(self):
        self.vector[self.process_id] += 1
        message_vector = self.vector.copy()
        print(
            f"{self.name}: Sending message -> "
            f"Vector = {message_vector}"
        )
        return message_vector
    def receive_message(self, received_vector):
        for i in range(len(self.vector)):
            self.vector[i] = max(
                self.vector[i],
                received_vector[i]
            )
        self.vector[self.process_id] += 1
        print(
            f"{self.name}: Received message "
            f"{received_vector} -> "
            f"Vector = {self.vector}"
        )
# Create three processes
P1 = VectorProcess("P1", 0, 3)
P2 = VectorProcess("P2", 1, 3)
P3 = VectorProcess("P3", 2, 3)
# P1 performs local event
P1.local_event()
# P1 sends message to P2
message = P1.send_message()
# P2 receives message
P2.receive_message(message)
# P2 performs local event
P2.local_event()
# P2 sends message to P3
message = P2.send_message()
# P3 receives message
P3.receive_message(message)
# P3 performs local event
P3.local_event()
```

SCREENSHOTS:

```
PS S:\sem 7\Cloud computing> python lamport_clock.py
P1: Local event -> Clock = 1
P1: Sending message -> Timestamp = 2
P2: Received message (Timestamp = 2) -> Clock = 3
P2: Local event -> Clock = 4
P2: Sending message -> Timestamp = 5
P1: Received message (Timestamp = 5) -> Clock = 6
```

```
PS S:\sem 7\Cloud computing> python vector_clock.py
P1: Local event -> Vector = [1, 0, 0]
P1: Sending message -> Vector = [2, 0, 0]
P2: Received message [2, 0, 0] -> Vector = [2, 1, 0]
P2: Local event -> Vector = [2, 2, 0]
P2: Sending message -> Vector = [2, 3, 0]
P3: Received message [2, 3, 0] -> Vector = [2, 3, 1]
P3: Local event -> Vector = [2, 3, 2]
```

OUTPUT:

Task 1 - Logical Clocks (Lamport's Logical Clock):

P1: Local event -> Clock = 1

P1: Sending message -> Timestamp = 2

P2: Received message (Timestamp = 2) -> Clock = 3

P2: Local event -> Clock = 4

P2: Sending message -> Timestamp = 5

P1: Received message (Timestamp = 5) -> Clock = 6

Task 2 - Vector Clocks:

Local event -> Vector = [1, 0, 0]

Sending message -> Vector = [2, 0, 0]

Received message [2, 0, 0] -> Vector = [2, 1, 0]

Local event -> Vector = [2, 2, 0]

Sending message -> Vector = [2, 3, 0]

Received message [2, 3, 0] -> Vector = [2, 3, 1]

Local event -> Vector = [2, 3, 2]

RESULT:

Successfully implemented logical and vector clocks, allowing consistent event ordering and causality detection across distributed processes.

Exercise: 05 Virtual Instances and Server Consolidation with Fault Tolerance in a Cloud

AIM:

To understand and implement the creation and management of virtual instances in a cloud environment, and to explore server consolidation strategies and implement basic fault-tolerance mechanisms to ensure high availability.

PROCEDURE:

Part 1: Virtual Instances Management

Task 1: Create and Configure Virtual Instances

- Create multiple virtual instances (VMs) in a cloud environment.
- Configure the virtual instances to run a simple web server.

Task 2: Implement Basic Load Balancing

- Set up a basic load balancer.
- Configure the load balancer to distribute incoming traffic among the virtual instances.

Part 2: Server Consolidation

Task 1: Implement Server Consolidation Strategy

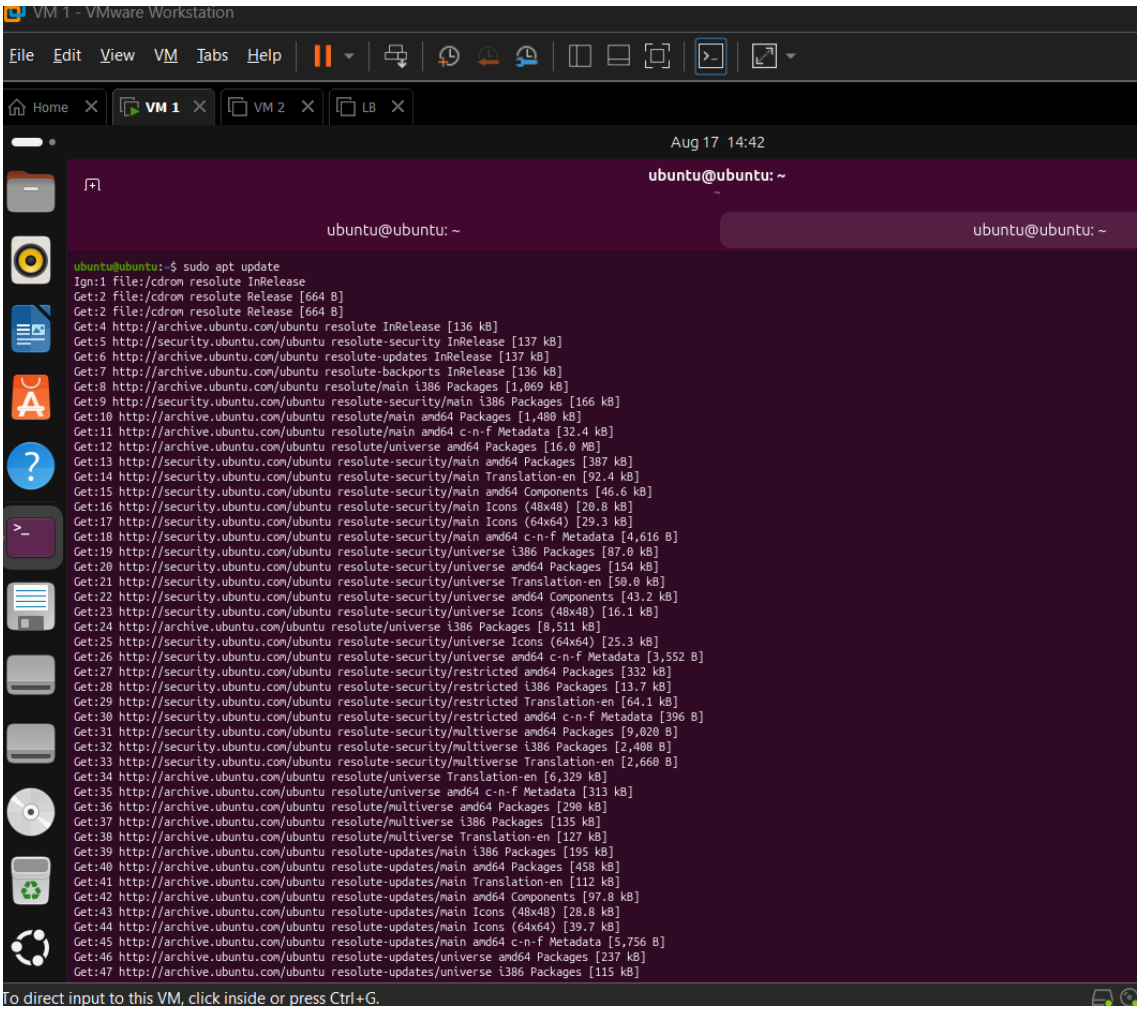
- Consolidate multiple VMs onto fewer physical servers or instances.
- Use the server consolidation strategy to optimize resource utilization and reduce costs.

Part 3: Fault Tolerance

Task 1: Implement Fault-Tolerant Design

- Design a fault-tolerant architecture to ensure high availability of services.
- Implement the fault-tolerant design.

SCREENSHOTS:



```
tched 38.5 MB in 9s (4,518 kB/s)
7 packages can be upgraded. Run 'apt list --upgradable' to see them.
tice: Skipping acquire of configured file 'restricted/binary-i386/Packages' as repository 'file:/cdrom resolute InRelease' doesn't sup
tice: Skipping acquire of configured file 'main/binary-i386/Packages' as repository 'file:/cdrom resolute InRelease' doesn't support a
untu@ubuntu:~$ sudo apt install apache2 -y
stalling:
apache2

stalling dependencies:
apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64

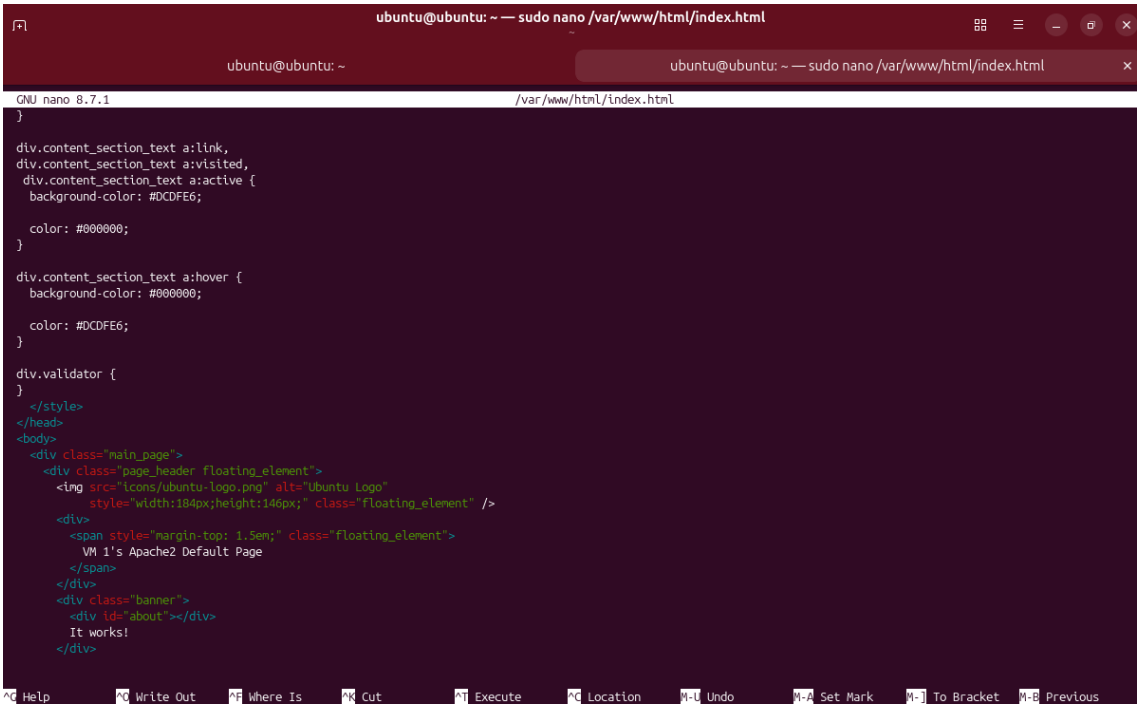
ggested packages:
apache2-doc  apache2-suexec-pristine  | apache2-suexec-custom

mmary:
Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 377
Download size: 1,938 kB
Space needed: 7,568 kB / 2,658 MB available

t:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libapr1t64 amd64 1.7.6-3 [109 kB]
t:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libaprutil1t64 amd64 1.6.3-3ubuntu3 [93.8 kB]
t:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-3ubuntu3 [11.4 kB]
t:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libaprutil1-ldap amd64 1.6.3-3ubuntu3 [9,226 B]
t:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-bin amd64 2.4.66-2ubuntu2.4 [1,355 kB]
t:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-data all 2.4.66-2ubuntu2.4 [163 kB]
t:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-utils amd64 2.4.66-2ubuntu2.4 [94.7 kB]
t:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2 amd64 2.4.66-2ubuntu2.4 [101 kB]
tched 1,938 kB in 3s (643 kB/s)
lecting previously unselected package libapr1t64:amd64.
eading database… 210055 files and directories currently installed.)
eparing to unpack …/0-libapr1t64_1.7.6-3_amd64.deb…
packing libapr1t64:amd64 (1.7.6-3)…
lecting previously unselected package libaprutil1t64:amd64.
eparing to unpack …/1-libaprutil1t64_1.6.3-3ubuntu3_amd64.deb…
packing libaprutil1t64:amd64 (1.6.3-3ubuntu3)…
lecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
eparing to unpack …/2-libaprutil1-dbd-sqlite3_1.6.3-3ubuntu3_amd64.deb…
packing libaprutil1-dbd-sqlite3:amd64 (1.6.3-3ubuntu3)…
lecting previously unselected package libaprutil1-ldap:amd64.
eparing to unpack …/3-libaprutil1-ldap_1.6.3-3ubuntu3_amd64.deb…
```

```
rocessing triggers for ufw (0.36.2-9build1)…
rocessing triggers for man-db (2.13.1-1build1)…
rocessing triggers for libc-bin (2.43-2ubuntu2)…
buntu@ubuntu:~$ sudo systemctl enable apache2
ynchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
xecuting: /usr/lib/systemd/systemd-sysv-install enable apache2
buntu@ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-08-17 14:40:55 UTC; 29s ago
Invocation: 4229250d8d3d4d1ba473e1849f5c181f
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 6976 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 55 (limit: 5345)
     Memory: 5.8M (peak: 6.1M)
        CPU: 105ms
     CGroup: /system.slice/apache2.service
             ├─6976 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─6992 /usr/sbin/apache2 -k start -DFOREGROUND
             └─6993 /usr/sbin/apache2 -k start -DFOREGROUND

ug 17 14:40:55 ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
ug 17 14:40:55 ubuntu apachectl[6976]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set
ug 17 14:40:55 ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
buntu@ubuntu:~$
```



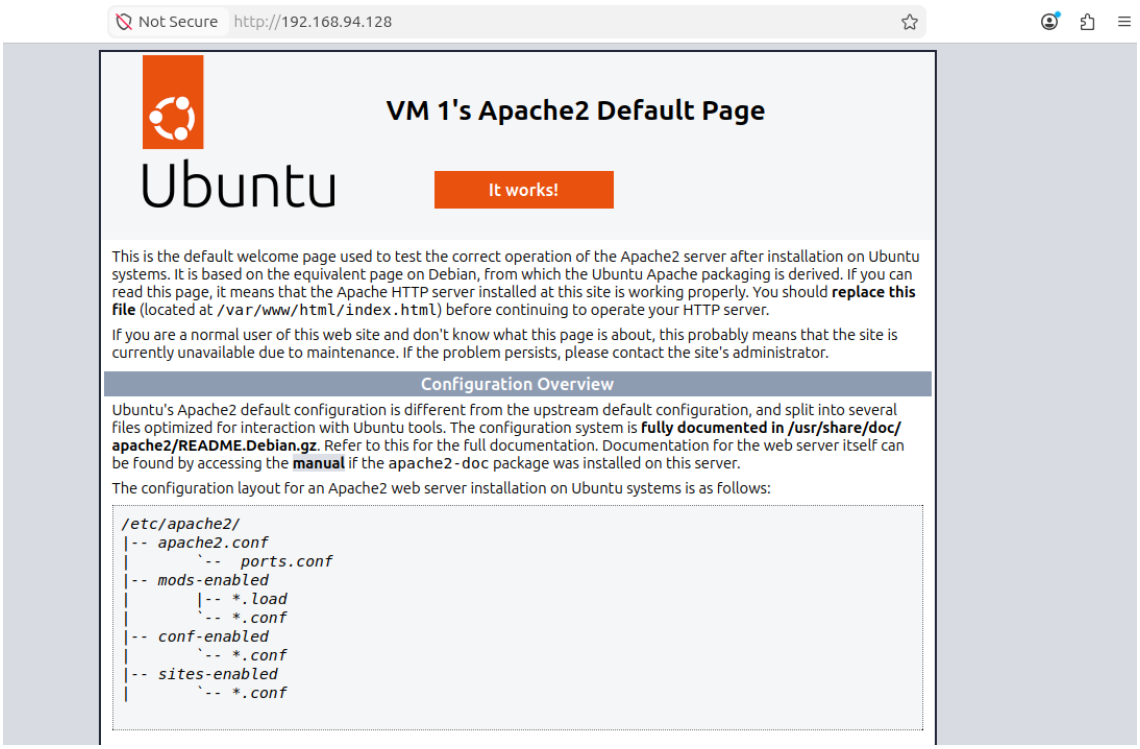

```
ubuntu@ubuntu:~$ sudo nano /var/www/html/index.html
ubuntu@ubuntu:~$ sudo apt install net-tools
Installing:
  net-tools

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 377
  Download size: 151 kB
  Space needed: 680 kB / 2,643 MB available

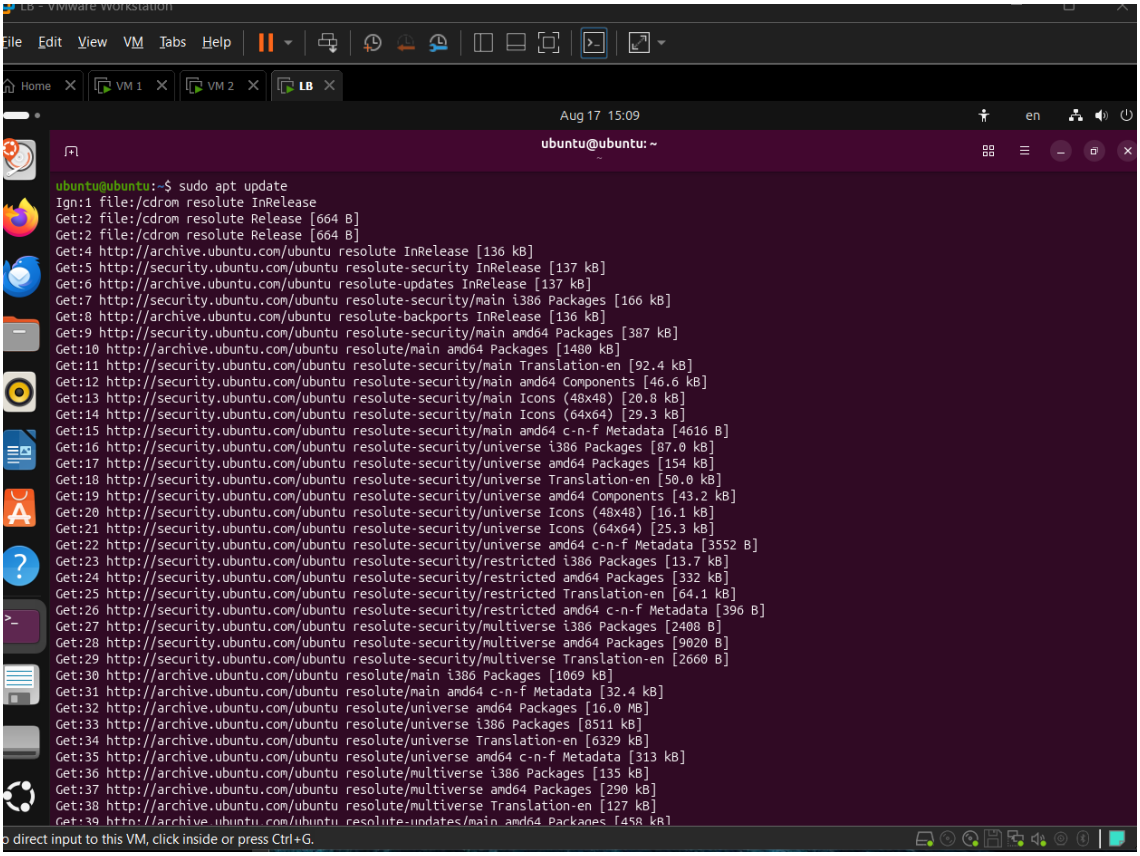
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1 [151 kB]
Fetched 151 kB in 2s (72.6 kB/s)
Selecting previously unselected package net-tools.
(Reading database… 210761 files and directories currently installed.)
Preparing to unpack …/net-tools_2.10-2ubuntu1_amd64.deb…
Unpacking net-tools (2.10-2ubuntu1)…
Setting up net-tools (2.10-2ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
ubuntu@ubuntu:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.94.128  netmask 255.255.255.0  broadcast 192.168.94.255
        inet6 fe80::20c:29ff:fe8d:a8d  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:8d:0a:8d  txqueuelen 1000  (Ethernet)
        RX packets 29333  bytes 42732179 (42.7 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 7399  bytes 529214 (529.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 504  bytes 43447 (43.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 504  bytes 43447 (43.4 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ubuntu@ubuntu:~$
```


Not Secure
http://192.168.94.128
VM 1's Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf


Not Secure
http://192.168.94.129
VM 2's Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf



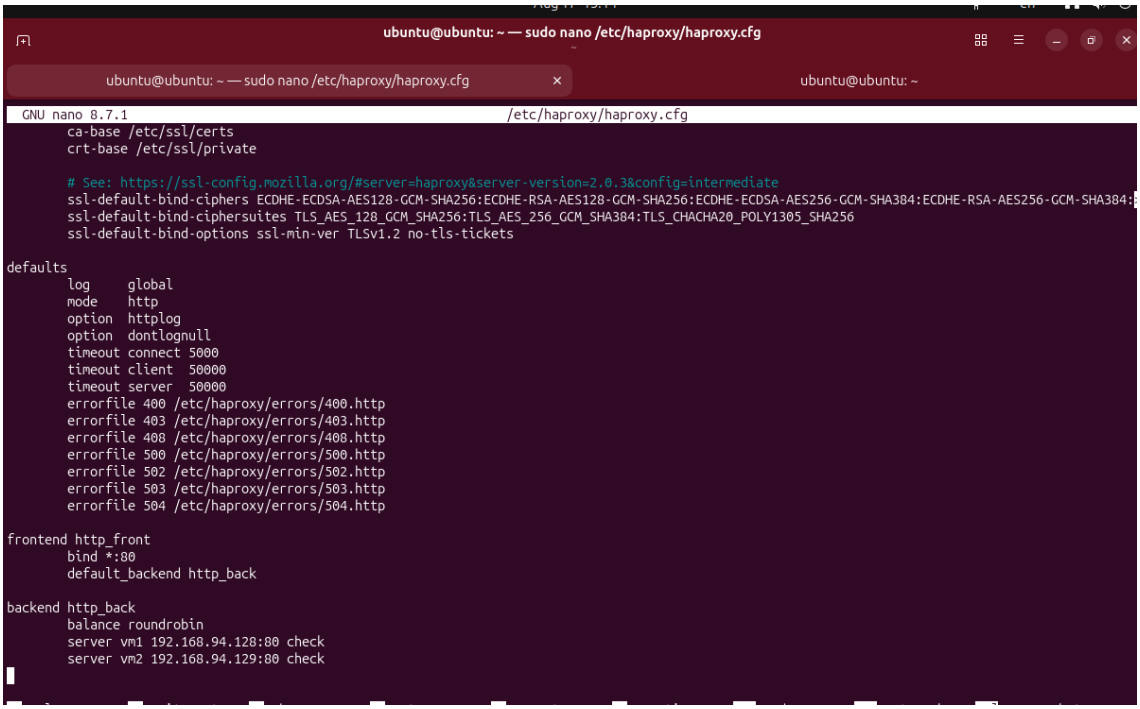

```
buntu@ubuntu:~$ sudo apt install haproxy -y
nstalling:
 haproxy

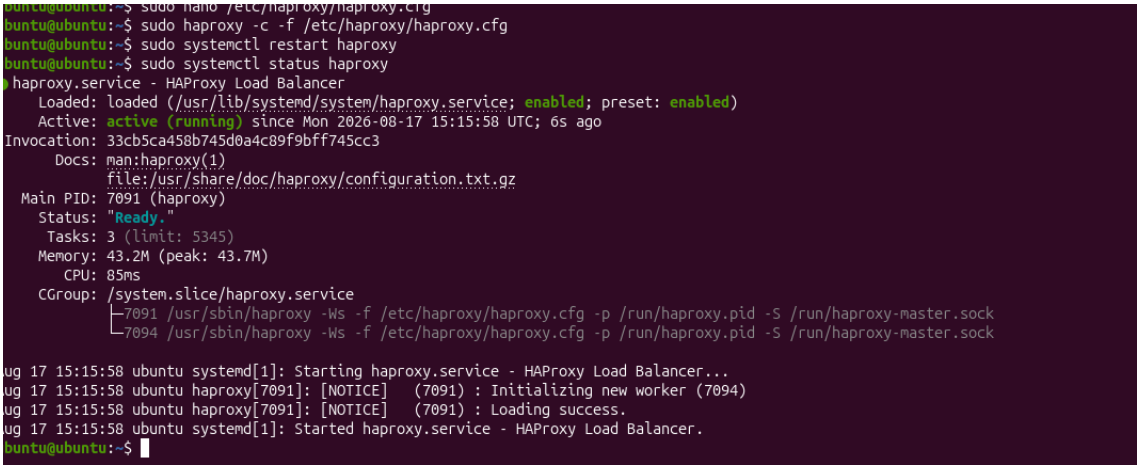
uggested packages:
 vim-haproxy  haproxy-doc

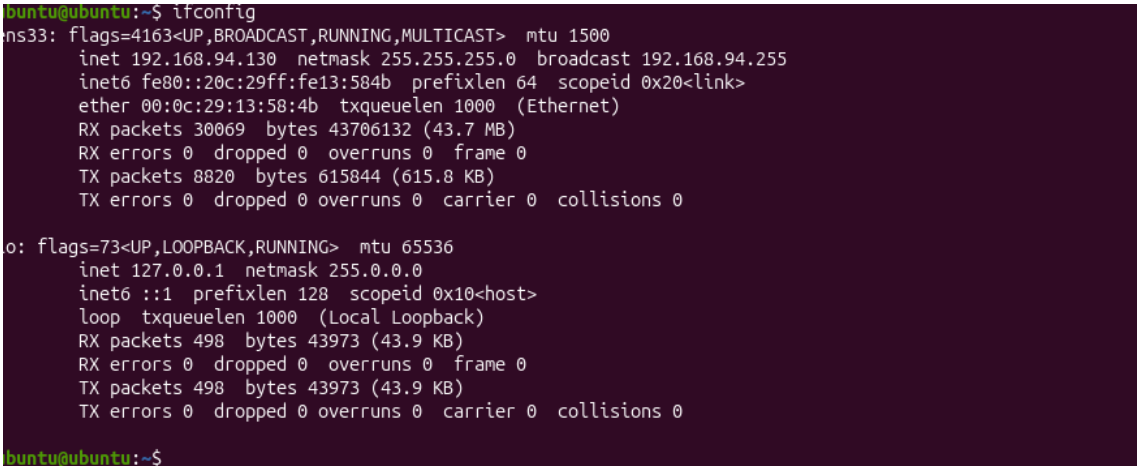
ummary:
 Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 377
 Download size: 2520 kB
 Space needed: 6022 kB / 2659 MB available

et:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 haproxy amd64 3.2.9-1ubuntu2.2 [2520 kB]
etched 2520 kB in 3s (768 kB/s)
electing previously unselected package haproxy.
Reading database ... 210055 files and directories currently installed.)
reparing to unpack .../haproxy_3.2.9-1ubuntu2.2_amd64.deb ...
npacking haproxy (3.2.9-1ubuntu2.2) ...
etting up haproxy (3.2.9-1ubuntu2.2) ...
reated symlink '/etc/systemd/system/multi-user.target.wants/haproxy.service' → '/usr/lib/systemd/system/haproxy.service'.
rocessing triggers for rsyslog (8.2512.0-1ubuntu4) ...
rocessing triggers for man-db (2.13.1-1build1) ...
buntu@ubuntu:~$
```

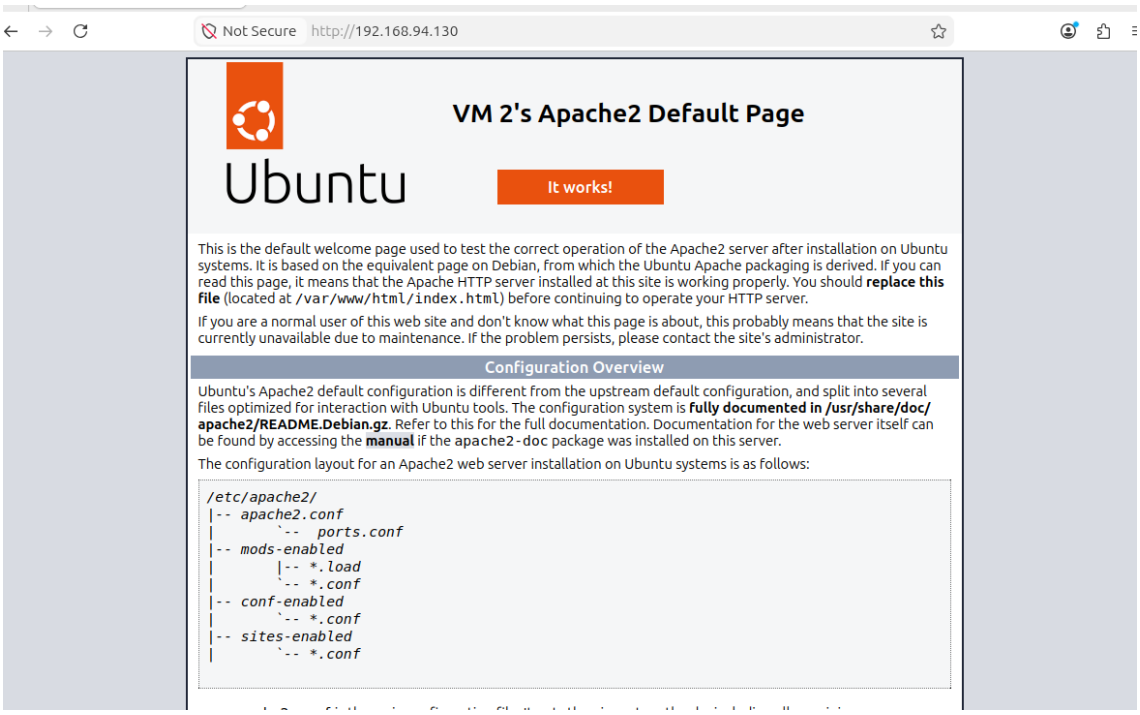
```
ubuntu@ubuntu:~$ sudo systemctl enable haproxy
Synchronizing state of haproxy.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable haproxy
ubuntu@ubuntu:~$ sudo cp /etc/haproxy/haproxy.cfg /etc/haproxy/haproxy.cfg.backup
ubuntu@ubuntu:~$ sudo nano /etc/haproxy/haproxy.cfg
ubuntu@ubuntu:~$
```

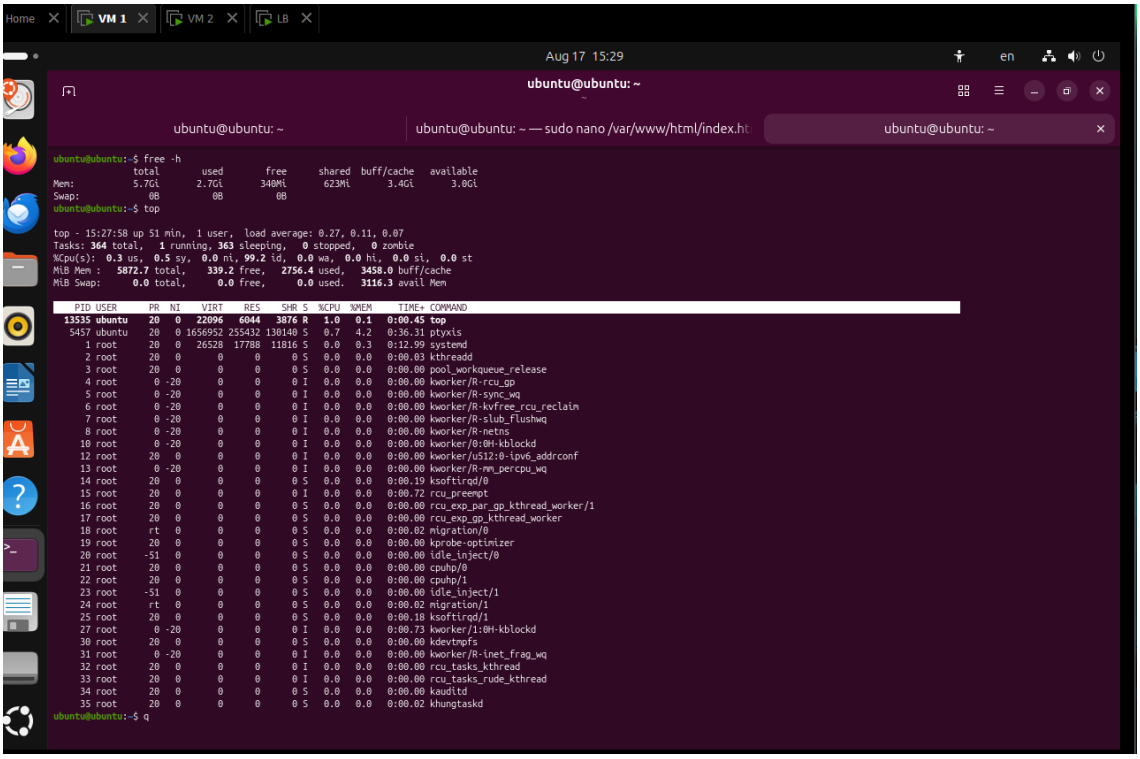

ubuntu@ubuntu: ~ — sudo nano /etc/haproxy/haproxy.cfg
ubuntu@ubuntu: ~ — sudo nano /etc/haproxy/haproxy.cfg
ubuntu@ubuntu: ~
GNU nano 8.7.1 /etc/haproxy/haproxy.cfg
ca-base /etc/ssl/certs
crt-base /etc/ssl/private
See: https://ssl-config.mozilla.org/#server=haproxy&server-version=2.0.3&config=intermediate
ssl-default-bind-ciphers ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-SHA384:
ssl-default-bind-ciphersuites TLS_AES_128_GCM_SHA256:TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256
ssl-default-bind-options ssl-min-ver TLSv1.2 no-tls-tickets
defaults
log global
mode http
option httplog
option dontlognull
timeout connect 5000
timeout client 50000
timeout server 50000
errorfile 400 /etc/haproxy/errors/400.http
errorfile 403 /etc/haproxy/errors/403.http
errorfile 408 /etc/haproxy/errors/408.http
errorfile 500 /etc/haproxy/errors/500.http
errorfile 502 /etc/haproxy/errors/502.http
errorfile 503 /etc/haproxy/errors/503.http
errorfile 504 /etc/haproxy/errors/504.http
frontend http_front
bind *:80
default_backend http_back
backend http_back
balance roundrobin
server vm1 192.168.94.128:80 check
server vm2 192.168.94.129:80 check

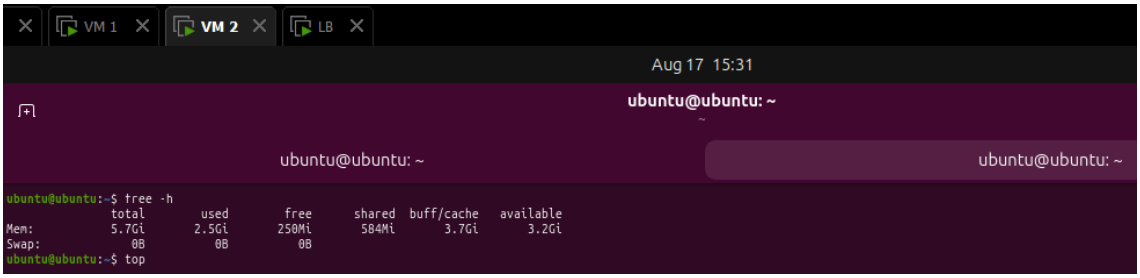

buntu@ubuntu:~$ sudo haproxy -c -f /etc/haproxy/haproxy.cfg
buntu@ubuntu:~$ sudo systemctl restart haproxy
buntu@ubuntu:~$ sudo systemctl status haproxy
haproxy.service - HAProxy Load Balancer
Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
Active: active (running) since Mon 2026-08-17 15:15:58 UTC; 6s ago
Invocation: 33cb5ca458b745d0a4c89f9bff745cc3
Docs: man:haproxy(1)
file:/usr/share/doc/haproxy/configuration.txt.gz
Main PID: 7091 (haproxy)
Status: "Ready."
Tasks: 3 (limit: 5345)
Memory: 43.2M (peak: 43.7M)
CPU: 85ms
CGroup: /system.slice/haproxy.service
├─7091 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
└─7094 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
ug 17 15:15:58 ubuntu systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
ug 17 15:15:58 ubuntu haproxy[7091]: [NOTICE] (7091) : Initializing new worker (7094)
ug 17 15:15:58 ubuntu haproxy[7091]: [NOTICE] (7091) : Loading success.
ug 17 15:15:58 ubuntu systemd[1]: Started haproxy.service - HAProxy Load Balancer.
buntu@ubuntu:~$

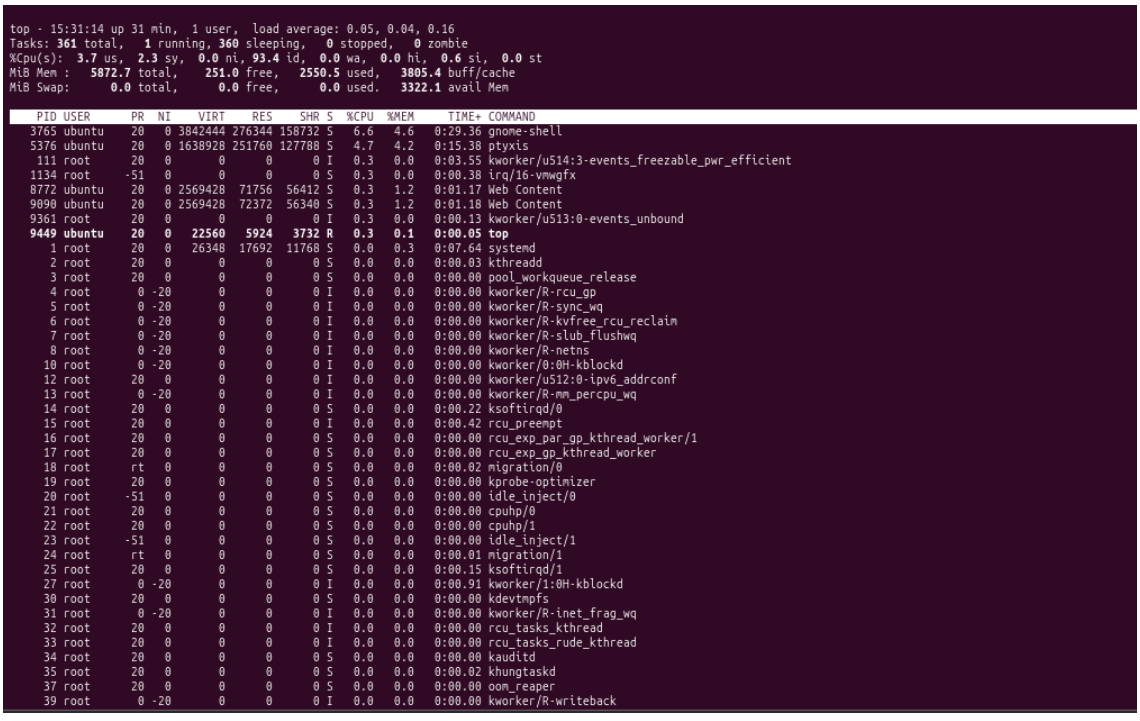

buntu@ubuntu:~$ ifconfig
ns33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.94.130 netmask 255.255.255.0 broadcast 192.168.94.255
inet6 fe80::20c:29ff:fe13:584b prefixlen 64 scopeid 0x20<link>
ether 00:0c:29:13:58:4b txqueuelen 1000 (Ethernet)
RX packets 30069 bytes 43706132 (43.7 MB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8820 bytes 615844 (615.8 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
o: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 498 bytes 43973 (43.9 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 498 bytes 43973 (43.9 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
buntu@ubuntu:~$


Apache2 Ubuntu Default Pa
Not Secure
http://192.168.94.130
VM 1's Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf


Not Secure
http://192.168.94.130
VM 2's Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf


Home
VM 1
VM 2
LB
Aug 17 15:29
en
ubuntu@ubuntu: ~
ubuntu@ubuntu: ~ — sudo nano /var/www/html/index.ht
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ free -h
total used free shared buff/cache available
Mem: 5.7Gi 2.7Gi 340Mi 623Mi 3.4Gi 3.0Gi
Swap: 0B 0B 0B
ubuntu@ubuntu:~$ top
top - 15:27:58 up 51 min, 1 user, load average: 0.27, 0.11, 0.07
Tasks: 364 total, 1 running, 363 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.5 sy, 0.0 ni, 99.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 5872.7 total, 339.2 free, 2756.4 used, 3458.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3116.3 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
13535 ubuntu 20 0 22096 6044 3876 R 1.0 0.1 0:00.45 top
5457 ubuntu 20 0 1656952 255432 130140 S 0.7 4.2 0:36.31 ptyxis
1 root 20 0 26528 17788 11816 S 0.0 0.3 0:12.99 systemd
2 root 20 0 0 0 0 S 0.0 0.0 0:00.03 kthreadd
3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_workqueue_release
4 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_gp
5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-sync_wq
6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-kvfree_rcu_reclaim
7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-slub_flushwq
8 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-netns
10 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/0:0H-kblockd
12 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/u512:0-ipv6_addrconf
13 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-mm_percpu_wq
14 root 20 0 0 0 0 S 0.0 0.0 0:00.19 ksoftirqd/0
15 root 20 0 0 0 0 I 0.0 0.0 0:00.72 rcu_preempt
16 root 20 0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_par_gp_kthread_worker/1
17 root 20 0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_gp_kthread_worker
18 root rt 0 0 0 0 S 0.0 0.0 0:00.02 migration/0
19 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kprobe-optimizer
20 root -51 0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/0
21 root 20 0 0 0 0 S 0.0 0.0 0:00.00 cpuhp/0
22 root 20 0 0 0 0 S 0.0 0.0 0:00.00 cpuhp/1
23 root -51 0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/1
24 root rt 0 0 0 0 S 0.0 0.0 0:00.02 migration/1
25 root 20 0 0 0 0 S 0.0 0.0 0:00.18 ksoftirqd/1
27 root 0 -20 0 0 0 I 0.0 0.0 0:00.73 kworker/1:0H-kblockd
30 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kdevtmpfs
31 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-inet_frag_wq
32 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_kthread
33 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_rude_kthread
34 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kauditd
35 root 20 0 0 0 0 S 0.0 0.0 0:00.02 khungtaskd
ubuntu@ubuntu:~$ q


VM 1
VM 2
LB
Aug 17 15:31
ubuntu@ubuntu: ~
ubuntu@ubuntu: ~
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ free -h
total used free shared buff/cache available
Mem: 5.7Gi 2.5Gi 250Mi 584Mi 3.7Gi 3.2Gi
Swap: 0B 0B 0B
ubuntu@ubuntu:~$ top


top - 15:31:14 up 31 min, 1 user, load average: 0.05, 0.04, 0.16
Tasks: 361 total, 1 running, 360 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.7 us, 2.3 sy, 0.0 ni, 93.4 id, 0.0 wa, 0.0 hi, 0.6 si, 0.0 st
MiB Mem : 5872.7 total, 251.0 free, 2550.5 used, 3805.4 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3322.1 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
3765 ubuntu 20 0 3842444 276344 158732 S 6.6 4.6 0:29.36 gnome-shell
5376 ubuntu 20 0 1638928 251760 127788 S 4.7 4.2 0:15.38 ptyxis
111 root 20 0 0 0 0 I 0.3 0.0 0:03.55 kworker/u514:3-events_freezable_pwr_efficient
1134 root -51 0 0 0 0 S 0.3 0.0 0:00.38 irq/16-vmwgfx
8772 ubuntu 20 0 2569428 71756 56412 S 0.3 1.2 0:01.17 Web Content
9090 ubuntu 20 0 2569428 72372 56340 S 0.3 1.2 0:01.18 Web Content
9361 root 20 0 0 0 0 I 0.3 0.0 0:00.13 kworker/u513:0-events_unbound
9449 ubuntu 20 0 22560 5924 3732 R 0.3 0.1 0:00.05 top
1 root 20 0 26348 17692 11768 S 0.0 0.3 0:07.64 systemd
2 root 20 0 0 0 0 S 0.0 0.0 0:00.03 kthreadd
3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_workqueue_release
4 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_gp
5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-sync_wq
6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-kvfree_rcu_reclaim
7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-slub_flushwq
8 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-netns
10 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/0:0H-kblockd
12 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/u512:0-ipv6_addrconf
13 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-mm_percpu_wq
14 root 20 0 0 0 0 S 0.0 0.0 0:00.22 ksoftirqd/0
15 root 20 0 0 0 0 I 0.0 0.0 0:00.42 rcu_preempt
16 root 20 0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_par_gp_kthread_worker/1
17 root 20 0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_gp_kthread_worker
18 root rt 0 0 0 0 S 0.0 0.0 0:00.02 migration/0
19 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kprobe-optimizer
20 root -51 0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/0
21 root 20 0 0 0 0 S 0.0 0.0 0:00.00 cpuhp/0
22 root 20 0 0 0 0 S 0.0 0.0 0:00.00 cpuhp/1
23 root -51 0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/1
24 root rt 0 0 0 0 S 0.0 0.0 0:00.01 migration/1
25 root 20 0 0 0 0 S 0.0 0.0 0:00.15 ksoftirqd/1
27 root 0 -20 0 0 0 I 0.0 0.0 0:00.91 kworker/1:0H-kblockd
30 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kdevtmpfs
31 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-inet_frag_wq
32 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_kthread
33 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_rude_kthread
34 root 20 0 0 0 0 S 0.0 0.0 0:00.00 kauditd
35 root 20 0 0 0 0 S 0.0 0.0 0:00.02 khungtaskd
37 root 20 0 0 0 0 S 0.0 0.0 0:00.00 oom_reaper
39 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-writeback

```
ubuntu@ubuntu:~$ sudo mkdir -p /var/www/html/vm2
```

```
ubuntu@ubuntu:~$ sudo cp /var/www/html/index.html /var/www/html/vm2/index.html
ubuntu@ubuntu:~$ sudo nano /var/www/html/vm2/index.html
ubuntu@ubuntu:~$
```

VM 1 | VM 2 | LB

Aug 17 15:43 en

ubuntu@ubuntu: ~ — sudo nano /var/www/html/vm2/index.html

ubuntu@ubuntu: ~ | ubuntu@ubuntu: ~ — sudo nano /var/www/html/index.ht | ubuntu@ubuntu: ~ — sudo nano /var/www/html/vm2/

GNU nano 8.7.1 /var/www/html/vm2/index.html

```
}

div.content_section_text a:link,
div.content_section_text a:visited,
 div.content_section_text a:active {
  background-color: #DCDFE6;

  color: #000000;
}

div.content_section_text a:hover {
  background-color: #000000;

  color: #DCDFE6;
}

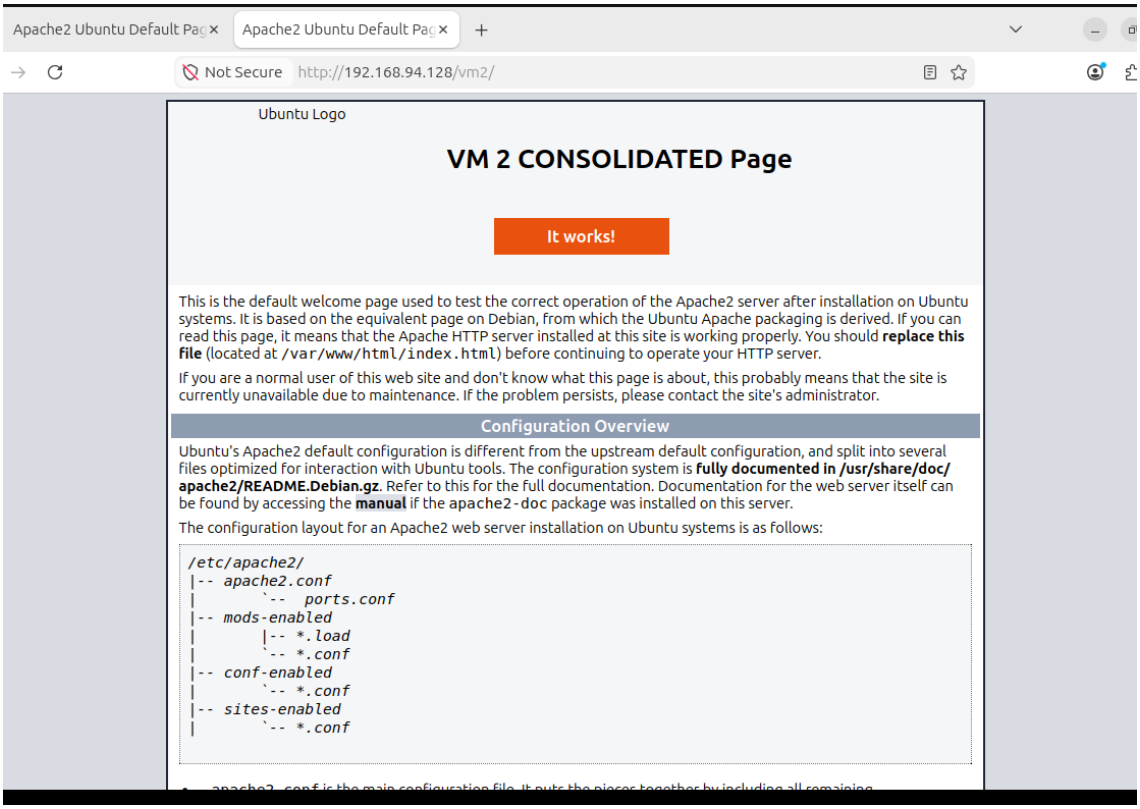
div.validator {
}
  </style>
</head>
<body>
  <div class="main_page">
    <div class="page_header floating_element">
      <img src="icons/ubuntu-logo.png" alt="Ubuntu Logo"
           style="width:184px;height:146px;" class="floating_element" />
      <div>
        <span style="margin-top: 1.5em;" class="floating_element">
          VM 2 CONSOLIDATED Apache2 Default Page
        </span>
      </div>
      <div class="banner">
        <div id="about"></div>
        It works!
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was

```
ubuntu@ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-08-17 15:48:29 UTC; 6s ago
 Invocation: dbb554015b334a61a4405f64320d6797
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 14131 (apache2)
     Status: "Processing requests..."
      Tasks: 55 (limit: 5345)
     Memory: 5.6M (peak: 5.7M)
        CPU: 97ms
     CGroup: /system.slice/apache2.service
             ├─14131 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─14134 /usr/sbin/apache2 -k start -DFOREGROUND
             └─14135 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 17 15:48:29 ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 17 15:48:29 ubuntu apachectl[14131]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Se
Aug 17 15:48:29 ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
```


Apache2 Ubuntu Default Pag
Apache2 Ubuntu Default Pag
Not Secure
http://192.168.94.128
VM 1's Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf


Apache2 Ubuntu Default Pag
Apache2 Ubuntu Default Pag
Not Secure
http://192.168.94.128/vm2/
Ubuntu Logo
VM 2 CONSOLIDATED Page
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf

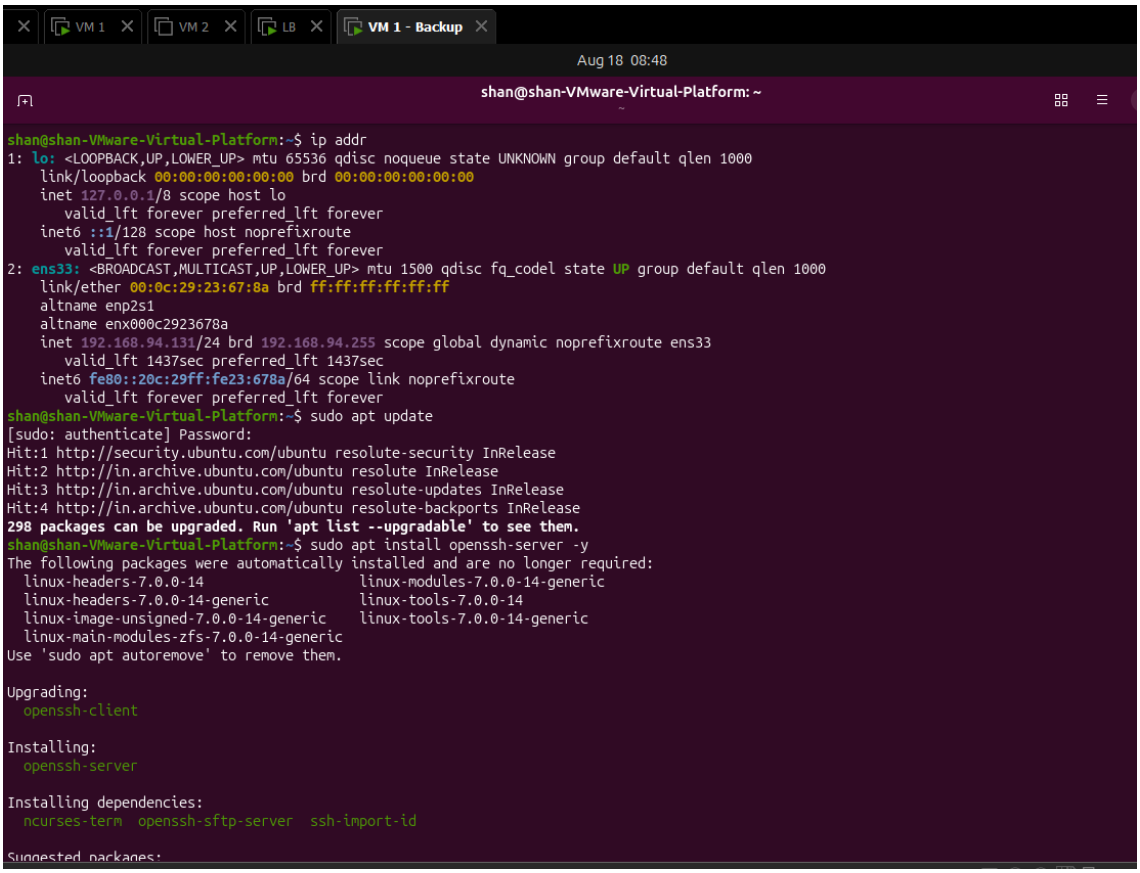
```
ubuntu@ubuntu:~$ sudo systemctl stop apache2
ubuntu@ubuntu:~$ sudo systemctl status apache2
○ apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: inactive (dead) since Mon 2026-08-17 15:56:12 UTC; 18s ago
   Duration: 53min 6.581s
 Invocation: d8b8d6aef6894af3b4df477f7fb57b14
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 6818 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Process: 9606 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 6818 (code=exited, status=0/SUCCESS)
     Status: "Total requests: 11; Idle/Busy workers 100/0;Requests/sec: 0.00346; Bytes served/sec:  10 B/sec"
   Mem peak: 9.2M
        CPU: 987ms

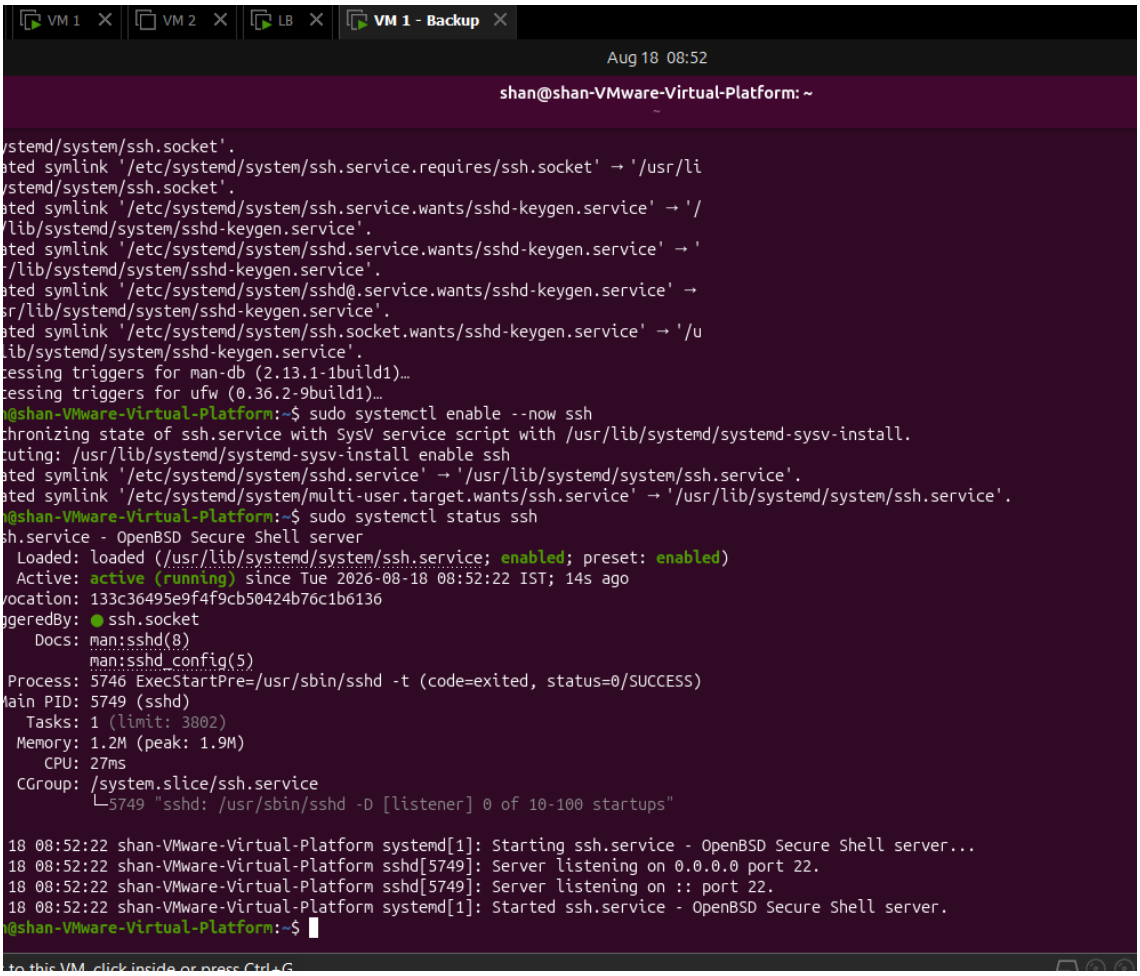
Aug 17 15:03:05 ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 17 15:03:05 ubuntu apachectl[6818]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName
Aug 17 15:03:05 ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
Aug 17 15:56:12 ubuntu systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Aug 17 15:56:12 ubuntu apachectl[9606]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName
Aug 17 15:56:12 ubuntu systemd[1]: apache2.service: Deactivated successfully.
Aug 17 15:56:12 ubuntu systemd[1]: Stopped apache2.service - The Apache HTTP Server.
ubuntu@ubuntu:~$
```

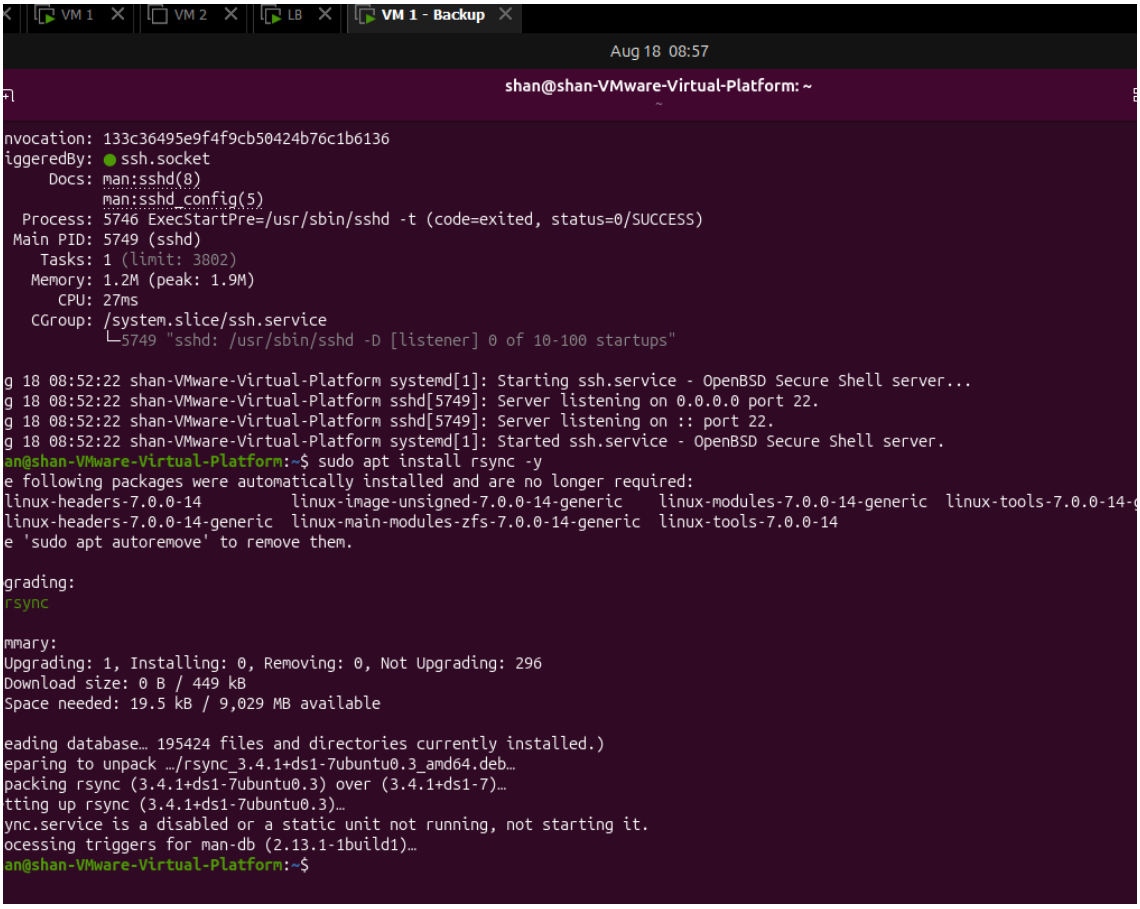
```
ubuntu@ubuntu:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           5.7Gi       2.7Gi       421Mi       647Mi       3.4Gi       3.1Gi
Swap:             0B          0B          0B
ubuntu@ubuntu:~$ top

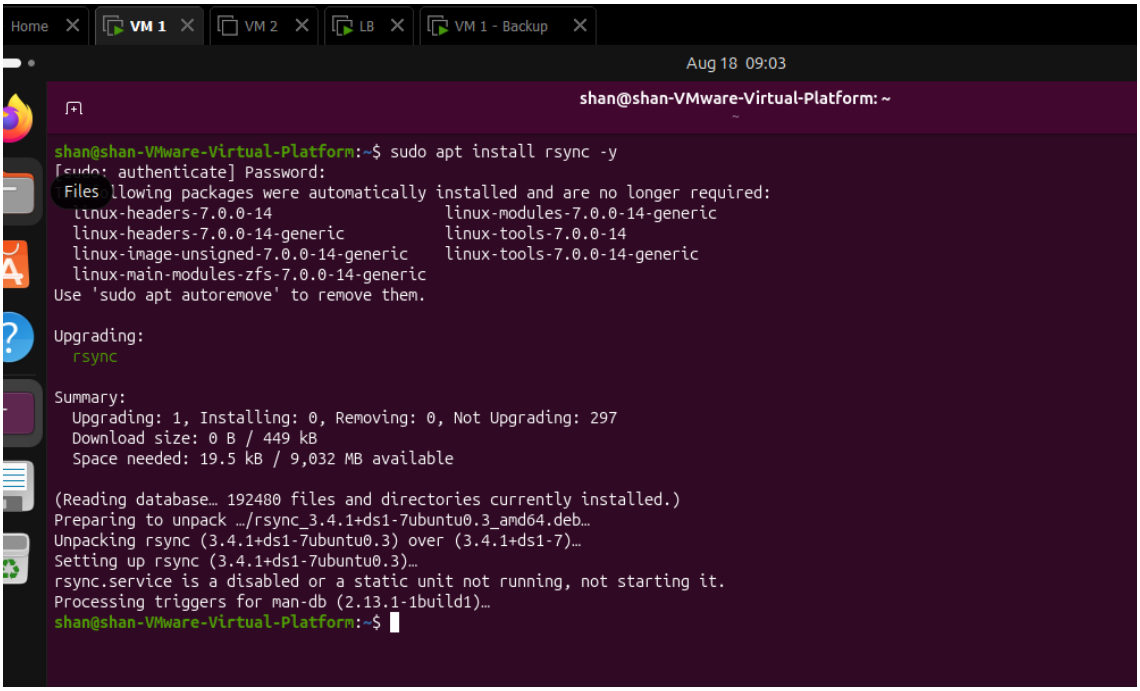
top - 15:58:07 up  1:21,  1 user,  load average: 0.14, 0.24, 0.19
Tasks: 370 total,   1 running, 369 sleeping,   0 stopped,   0 zombie
%Cpu(s): 16.0 us, 16.0 sy,  0.0 ni, 68.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   5872.7 total,    421.2 free,   2727.3 used,   3489.6 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   3145.4 avail Mem

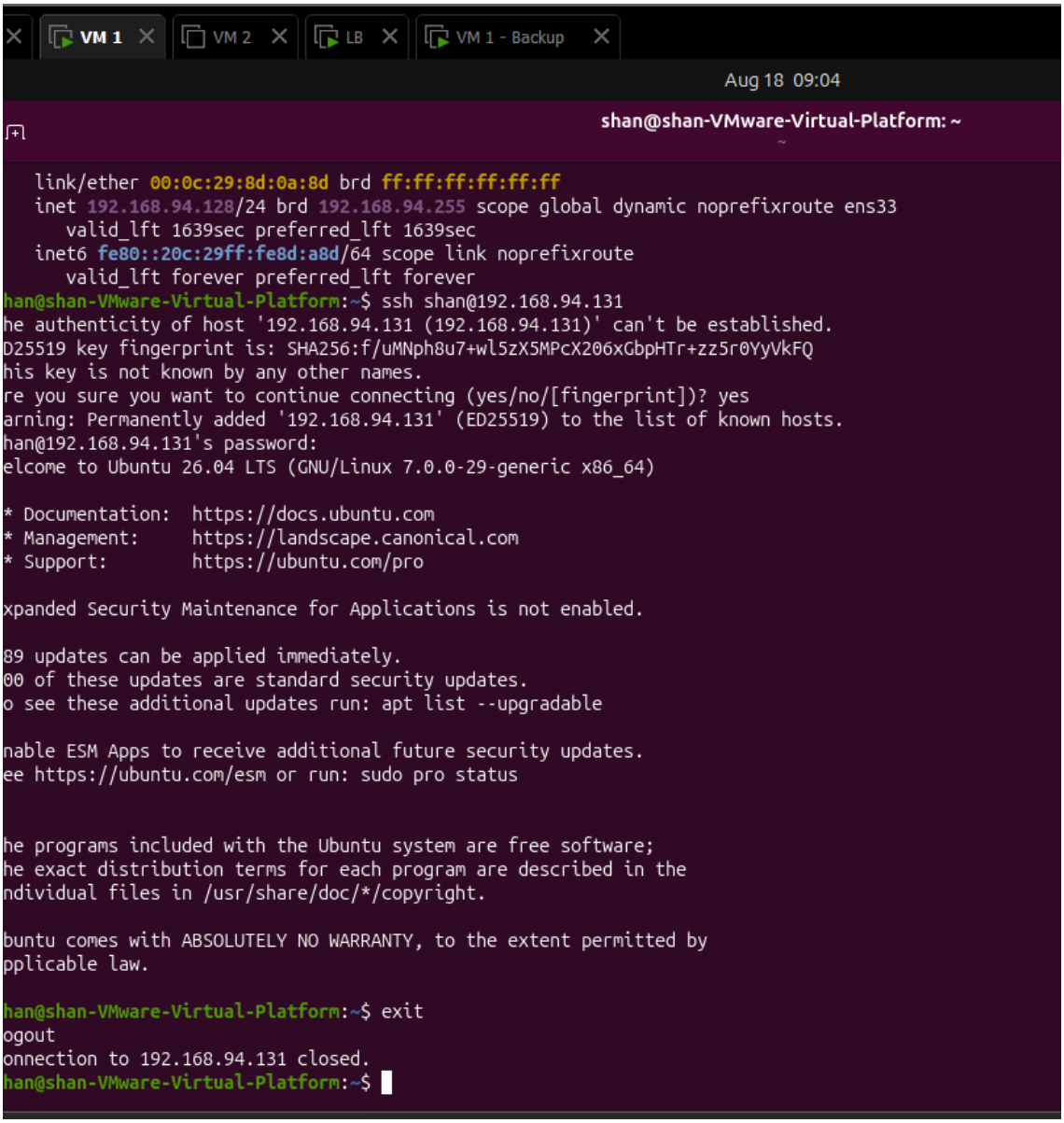
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   5457 ubuntu    20   0 1659740 255432 130140 S  15.4   4.2   1:07.48 ptyxis
   3786 ubuntu    20   0 3910116 281240 160460 S   7.7   4.7   2:26.51 gnome-shell
  14421 ubuntu    20   0   22092   5624   3476 R   7.7   0.1   0:00.03 top
      1 root      20   0   26528  17792  11816 S   0.0   0.3   0:13.54 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.04 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u512:0-ipv6_addrconf
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.22 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:01.08 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/1
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.04 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kprobe-optimizer
     20 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
```

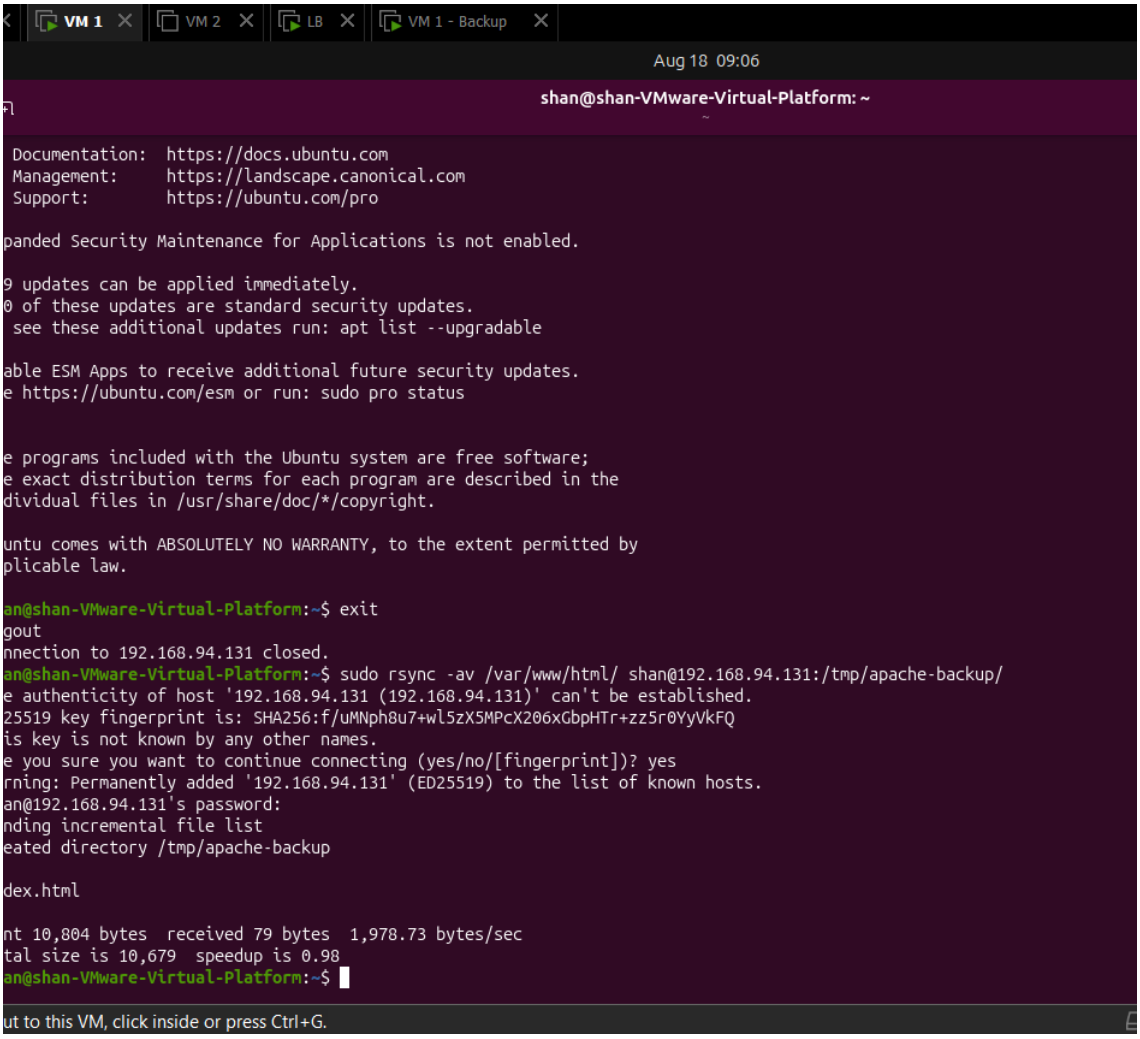

VM 1
VM 2
LB
VM 1 - Backup
Aug 18 08:48
shan@shan-VMware-Virtual-Platform: ~
shan@shan-VMware-Virtual-Platform:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:23:67:8a brd ff:ff:ff:ff:ff:ff
altname enp2s1
altname enx000c2923678a
inet 192.168.94.131/24 brd 192.168.94.255 scope global dynamic noprefixroute ens33
valid_lft 1437sec preferred_lft 1437sec
inet6 fe80::20c:29ff:fe23:678a/64 scope link noprefixroute
valid_lft forever preferred_lft forever
shan@shan-VMware-Virtual-Platform:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
298 packages can be upgraded. Run 'apt list --upgradable' to see them.
shan@shan-VMware-Virtual-Platform:~$ sudo apt install openssh-server -y
The following packages were automatically installed and are no longer required:
linux-headers-7.0.0-14
linux-modules-7.0.0-14-generic
linux-headers-7.0.0-14-generic
linux-tools-7.0.0-14
linux-image-unsigned-7.0.0-14-generic
linux-tools-7.0.0-14-generic
linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.
Upgrading:
openssh-client
Installing:
openssh-server
Installing dependencies:
ncurses-term
openssh-sftp-server
ssh-import-id
Suggested packages:

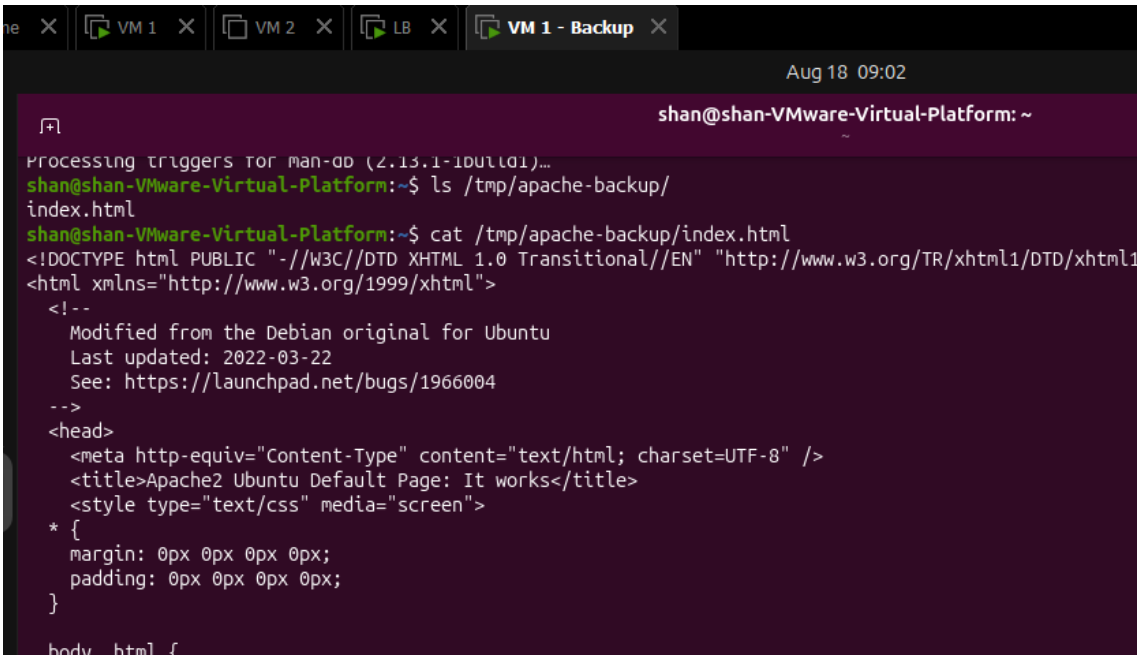

VM 1
VM 2
LB
VM 1 - Backup
Aug 18 08:52
shan@shan-VMware-Virtual-Platform: ~
ystemd/system/ssh.socket'.
ated symlink '/etc/systemd/system/ssh.service.requires/ssh.socket' → '/usr/li
ystemd/system/ssh.socket'.
ated symlink '/etc/systemd/system/ssh.service.wants/sshd-keygen.service' → '/
/lib/systemd/system/sshd-keygen.service'.
ated symlink '/etc/systemd/system/sshd.service.wants/sshd-keygen.service' → '
r/lib/systemd/system/sshd-keygen.service'.
ated symlink '/etc/systemd/system/sshd@.service.wants/sshd-keygen.service' →
sr/lib/systemd/system/sshd-keygen.service'.
ated symlink '/etc/systemd/system/ssh.socket.wants/sshd-keygen.service' → '/u
lib/systemd/system/sshd-keygen.service'.
cessing triggers for man-db (2.13.1-1build1)…
cessing triggers for ufw (0.36.2-9build1)…
@shan-VMware-Virtual-Platform:~$ sudo systemctl enable --now ssh
chronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
cuting: /usr/lib/systemd/systemd-sysv-install enable ssh
ated symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.
ated symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
@shan-VMware-Virtual-Platform:~$ sudo systemctl status ssh
sh.service - OpenBSD Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-08-18 08:52:22 IST; 14s ago
vocation: 133c36495e9f4f9cb50424b76c1b6136
ggeredBy: ● ssh.socket
Docs: man:sshd(8)
man:sshd_config(5)
Process: 5746 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
Main PID: 5749 (sshd)
Tasks: 1 (limit: 3802)
Memory: 1.2M (peak: 1.9M)
CPU: 27ms
CGroup: /system.slice/ssh.service
└─5749 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
18 08:52:22 shan-VMware-Virtual-Platform systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
18 08:52:22 shan-VMware-Virtual-Platform sshd[5749]: Server listening on 0.0.0.0 port 22.
18 08:52:22 shan-VMware-Virtual-Platform sshd[5749]: Server listening on :: port 22.
18 08:52:22 shan-VMware-Virtual-Platform systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
@shan-VMware-Virtual-Platform:~$
to this VM, click inside or press Ctrl+G

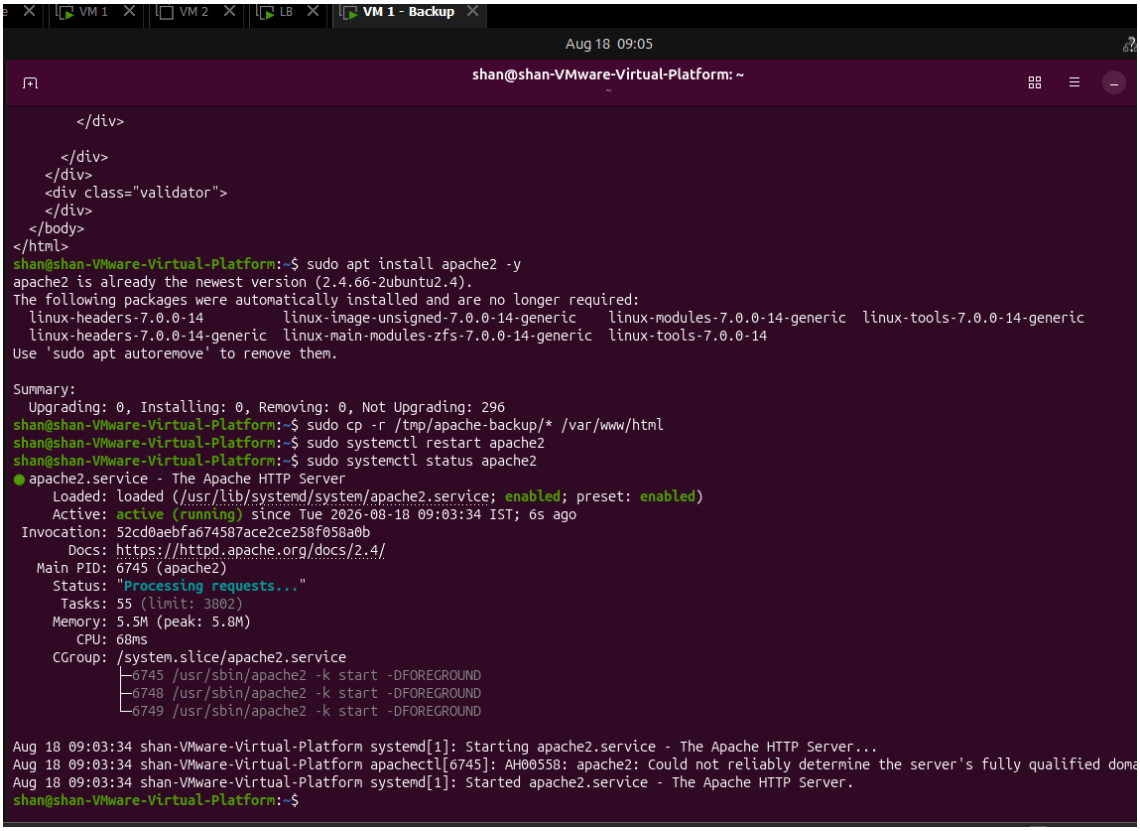

VM 1
VM 2
LB
VM 1 - Backup
Aug 18 08:57
shan@shan-VMware-Virtual-Platform: ~
nvocation: 133c36495e9f4f9cb50424b76c1b6136
iggeredBy: ● ssh.socket
Docs: man:sshd(8)
man:sshd_config(5)
Process: 5746 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
Main PID: 5749 (sshd)
Tasks: 1 (limit: 3802)
Memory: 1.2M (peak: 1.9M)
CPU: 27ms
CGroup: /system.slice/ssh.service
└5749 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
g 18 08:52:22 shan-VMware-Virtual-Platform systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
g 18 08:52:22 shan-VMware-Virtual-Platform sshd[5749]: Server listening on 0.0.0.0 port 22.
g 18 08:52:22 shan-VMware-Virtual-Platform sshd[5749]: Server listening on :: port 22.
g 18 08:52:22 shan-VMware-Virtual-Platform systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
an@shan-VMware-Virtual-Platform:~$ sudo apt install rsync -y
e following packages were automatically installed and are no longer required:
linux-headers-7.0.0-14 linux-image-unsigned-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-
linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14
e 'sudo apt autoremove' to remove them.
grading:
rsync
mmary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 296
Download size: 0 B / 449 kB
Space needed: 19.5 kB / 9,029 MB available
eading database… 195424 files and directories currently installed.)
eparing to unpack …/rsync_3.4.1+ds1-7ubuntu0.3_amd64.deb…
packing rsync (3.4.1+ds1-7ubuntu0.3) over (3.4.1+ds1-7)…
tting up rsync (3.4.1+ds1-7ubuntu0.3)…
ync.service is a disabled or a static unit not running, not starting it.
ocessing triggers for man-db (2.13.1-1build1)…
an@shan-VMware-Virtual-Platform:~$


Home
VM 1
VM 2
LB
VM 1 - Backup
Aug 18 09:03
shan@shan-VMware-Virtual-Platform: ~
Files
shan@shan-VMware-Virtual-Platform:~$ sudo apt install rsync -y
[sudo: authenticate] Password:
llowing packages were automatically installed and are no longer required:
linux-headers-7.0.0-14
linux-headers-7.0.0-14-generic
linux-image-unsigned-7.0.0-14-generic
linux-main-modules-zfs-7.0.0-14-generic
linux-modules-7.0.0-14-generic
linux-tools-7.0.0-14
linux-tools-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.
Upgrading:
rsync
Summary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 297
Download size: 0 B / 449 kB
Space needed: 19.5 kB / 9,032 MB available
(Reading database… 192480 files and directories currently installed.)
Preparing to unpack …/rsync_3.4.1+ds1-7ubuntu0.3_amd64.deb…
Unpacking rsync (3.4.1+ds1-7ubuntu0.3) over (3.4.1+ds1-7)…
Setting up rsync (3.4.1+ds1-7ubuntu0.3)…
rsync.service is a disabled or a static unit not running, not starting it.
Processing triggers for man-db (2.13.1-1build1)…
shan@shan-VMware-Virtual-Platform:~$

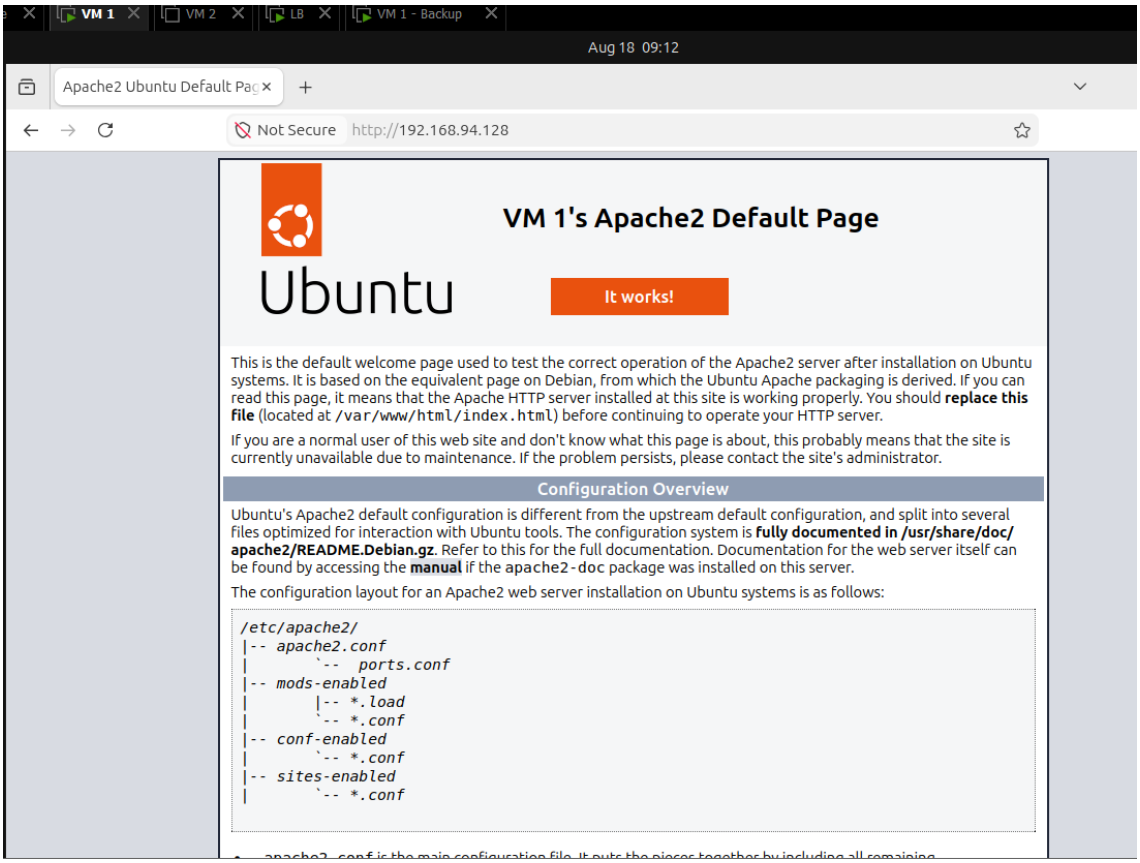

VM 1
VM 2
LB
VM 1 - Backup
Aug 18 09:04
shan@shan-VMware-Virtual-Platform: ~
link/ether 00:0c:29:8d:0a:8d brd ff:ff:ff:ff:ff:ff
inet 192.168.94.128/24 brd 192.168.94.255 scope global dynamic noprefixroute ens33
valid_lft 1639sec preferred_lft 1639sec
inet6 fe80::20c:29ff:fe8d:a8d/64 scope link noprefixroute
valid_lft forever preferred_lft forever
han@shan-VMware-Virtual-Platform:~$ ssh shan@192.168.94.131
he authenticity of host '192.168.94.131 (192.168.94.131)' can't be established.
D25519 key fingerprint is: SHA256:f/uMNph8u7+wl5zX5MPcX206xGbpHTr+zz5r0YyVkFQ
his key is not known by any other names.
re you sure you want to continue connecting (yes/no/[fingerprint])? yes
arning: Permanently added '192.168.94.131' (ED25519) to the list of known hosts.
han@192.168.94.131's password:
elcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
xpanded Security Maintenance for Applications is not enabled.
89 updates can be applied immediately.
00 of these updates are standard security updates.
o see these additional updates run: apt list --upgradable
nable ESM Apps to receive additional future security updates.
ee https://ubuntu.com/esm or run: sudo pro status
he programs included with the Ubuntu system are free software;
he exact distribution terms for each program are described in the
ndividual files in /usr/share/doc/*/copyright.
buntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
pplicable law.
han@shan-VMware-Virtual-Platform:~$ exit
ogout
onnection to 192.168.94.131 closed.
han@shan-VMware-Virtual-Platform:~$

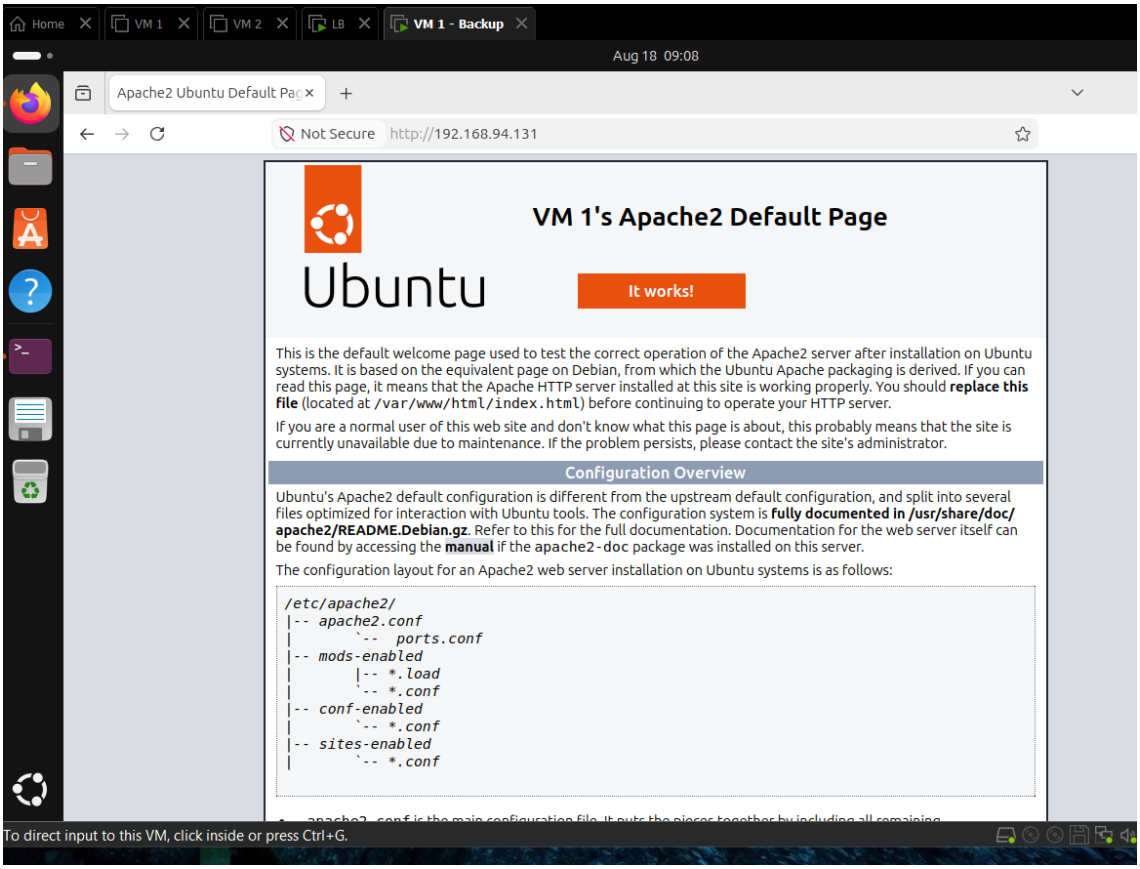

VM 1
VM 2
LB
VM 1 - Backup
Aug 18 09:06
shan@shan-VMware-Virtual-Platform: ~
Documentation: https://docs.ubuntu.com
Management: https://landscape.canonical.com
Support: https://ubuntu.com/pro
panded Security Maintenance for Applications is not enabled.
9 updates can be applied immediately.
0 of these updates are standard security updates.
see these additional updates run: apt list --upgradable
able ESM Apps to receive additional future security updates.
e https://ubuntu.com/esm or run: sudo pro status
e programs included with the Ubuntu system are free software;
e exact distribution terms for each program are described in the
dividual files in /usr/share/doc/*/copyright.
untu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
plicable law.
an@shan-VMware-Virtual-Platform:~$ exit
gout
nnection to 192.168.94.131 closed.
an@shan-VMware-Virtual-Platform:~$ sudo rsync -av /var/www/html/ shan@192.168.94.131:/tmp/apache-backup/
e authenticity of host '192.168.94.131 (192.168.94.131)' can't be established.
25519 key fingerprint is: SHA256:f/uMNph8u7+wl5zX5MPcX206xGbpHTr+zz5r0YyVkFQ
is key is not known by any other names.
e you sure you want to continue connecting (yes/no/[fingerprint])? yes
rning: Permanently added '192.168.94.131' (ED25519) to the list of known hosts.
an@192.168.94.131's password:
nding incremental file list
eated directory /tmp/apache-backup
dex.html
nt 10,804 bytes received 79 bytes 1,978.73 bytes/sec
tal size is 10,679 speedup is 0.98
an@shan-VMware-Virtual-Platform:~$
ut to this VM, click inside or press Ctrl+G.


VM 1
VM 2
LB
VM 1 - Backup
Aug 18 09:02
shan@shan-VMware-Virtual-Platform: ~
Processing triggers for man-db (2.13.1-1build1)…
shan@shan-VMware-Virtual-Platform:~$ ls /tmp/apache-backup/
index.html
shan@shan-VMware-Virtual-Platform:~$ cat /tmp/apache-backup/index.html
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1
<html xmlns="http://www.w3.org/1999/xhtml">
<!--
Modified from the Debian original for Ubuntu
Last updated: 2022-03-22
See: https://launchpad.net/bugs/1966004
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
<title>Apache2 Ubuntu Default Page: It works</title>
<style type="text/css" media="screen">
* {
margin: 0px 0px 0px 0px;
padding: 0px 0px 0px 0px;
}
body, html {


VM 1
VM 2
LB
VM 1 - Backup
Aug 18 09:05
shan@shan-VMware-Virtual-Platform: ~

</body>
</html>
shan@shan-VMware-Virtual-Platform:~$ sudo apt install apache2 -y
apache2 is already the newest version (2.4.66-2ubuntu2.4).
The following packages were automatically installed and are no longer required:
linux-headers-7.0.0-14 linux-image-unsigned-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-generic
linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 296
shan@shan-VMware-Virtual-Platform:~$ sudo cp -r /tmp/apache-backup/* /var/www/html
shan@shan-VMware-Virtual-Platform:~$ sudo systemctl restart apache2
shan@shan-VMware-Virtual-Platform:~$ sudo systemctl status apache2
apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-08-18 09:03:34 IST; 6s ago
Invocation: 52cd0aebfa674587ace2ce258f058a0b
Docs: https://httpd.apache.org/docs/2.4/
Main PID: 6745 (apache2)
Status: "Processing requests..."
Tasks: 55 (limit: 3802)
Memory: 5.5M (peak: 5.8M)
CPU: 68ms
CGroup: /system.slice/apache2.service
6745 /usr/sbin/apache2 -k start -DFOREGROUND
6748 /usr/sbin/apache2 -k start -DFOREGROUND
6749 /usr/sbin/apache2 -k start -DFOREGROUND
Aug 18 09:03:34 shan-VMware-Virtual-Platform systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 18 09:03:34 shan-VMware-Virtual-Platform apachectl[6745]: AH00558: apache2: Could not reliably determine the server's fully qualified doma
Aug 18 09:03:34 shan-VMware-Virtual-Platform systemd[1]: Started apache2.service - The Apache HTTP Server.
shan@shan-VMware-Virtual-Platform:~$


VM 1
VM 2
LB
VM 1 - Backup
Aug 18 09:12
Apache2 Ubuntu Default Pag
Not Secure
http://192.168.94.128
VM 1's Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf



OUTPUT:

VM 1 and VM 2 display the Apache2 Ubuntu Default Page.

HAProxy service status: Active: active (running).

VM 2 CONSOLIDATED Page is displayed.

VM 1 - Backup displays the VM 1 Apache2 Ubuntu Default Page.

VM 1 is not running and VM 1 - Backup shows the web page of VM 1.

RESULT:

Virtual instances with load balancing, server consolidation, and fault tolerance were successfully deployed, optimizing performance and ensuring service continuity.

Exercise 6: Task Scheduling and Load Balancing in a Cloud Environment

AIM:

To automate task scheduling and implement load balancing in a cloud environment for efficient resource management and high availability.

PROCEDURE**:**

Part 1: Task Scheduling

Task 1: Implement Task Scheduling with Cloud Services

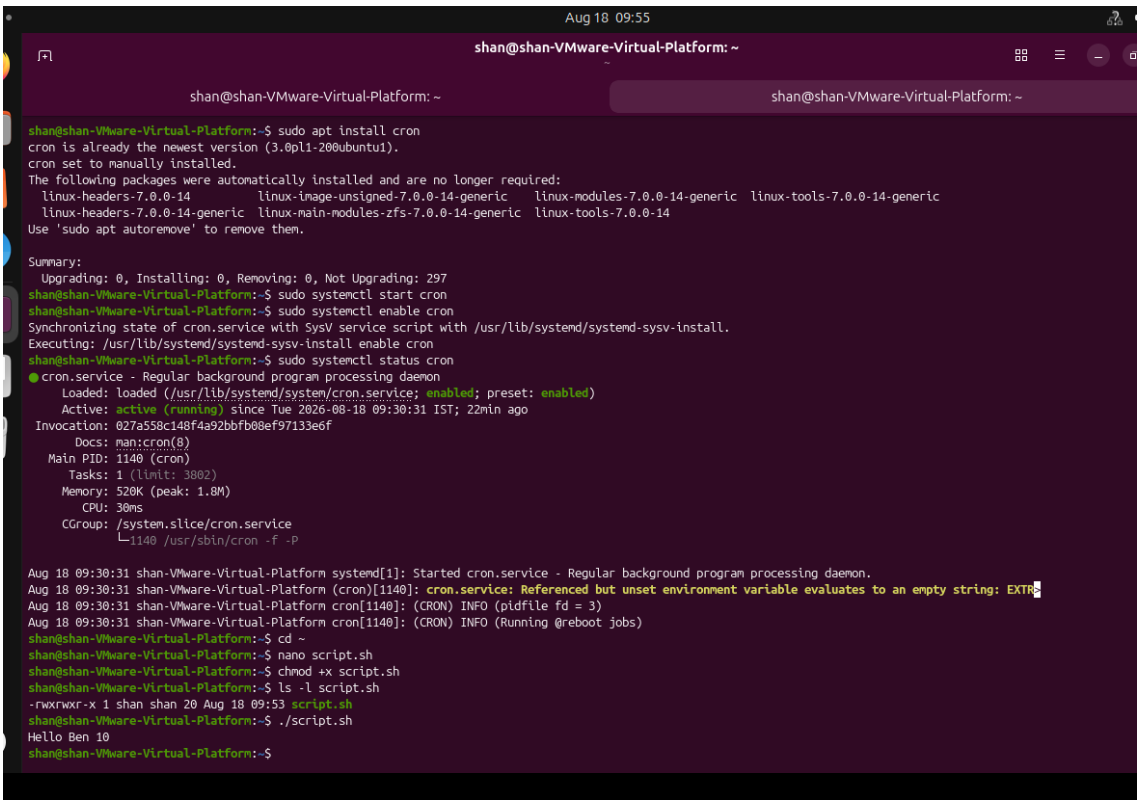
- Implement task scheduling in the cloud environment.
- Use the scheduling mechanism to automate task execution for efficient resource management and high availability.

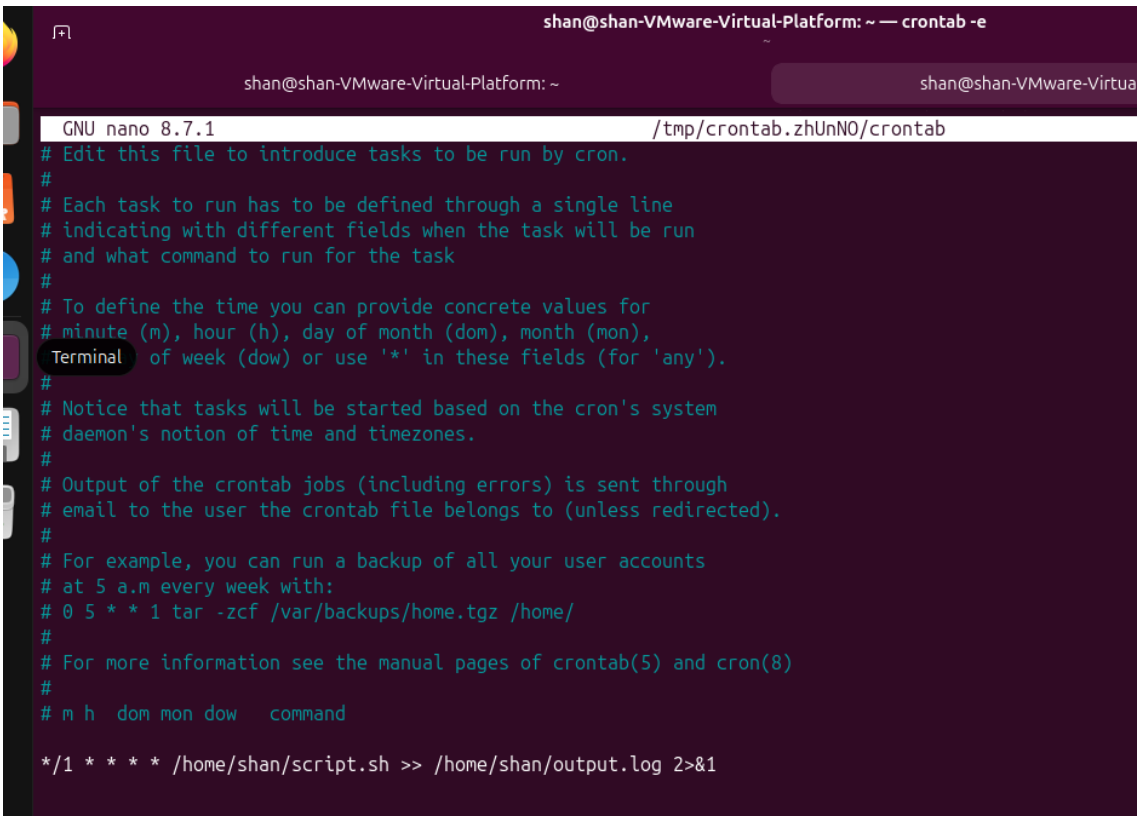
Part 2: Advanced Scheduling and Load Balancing (Optional)

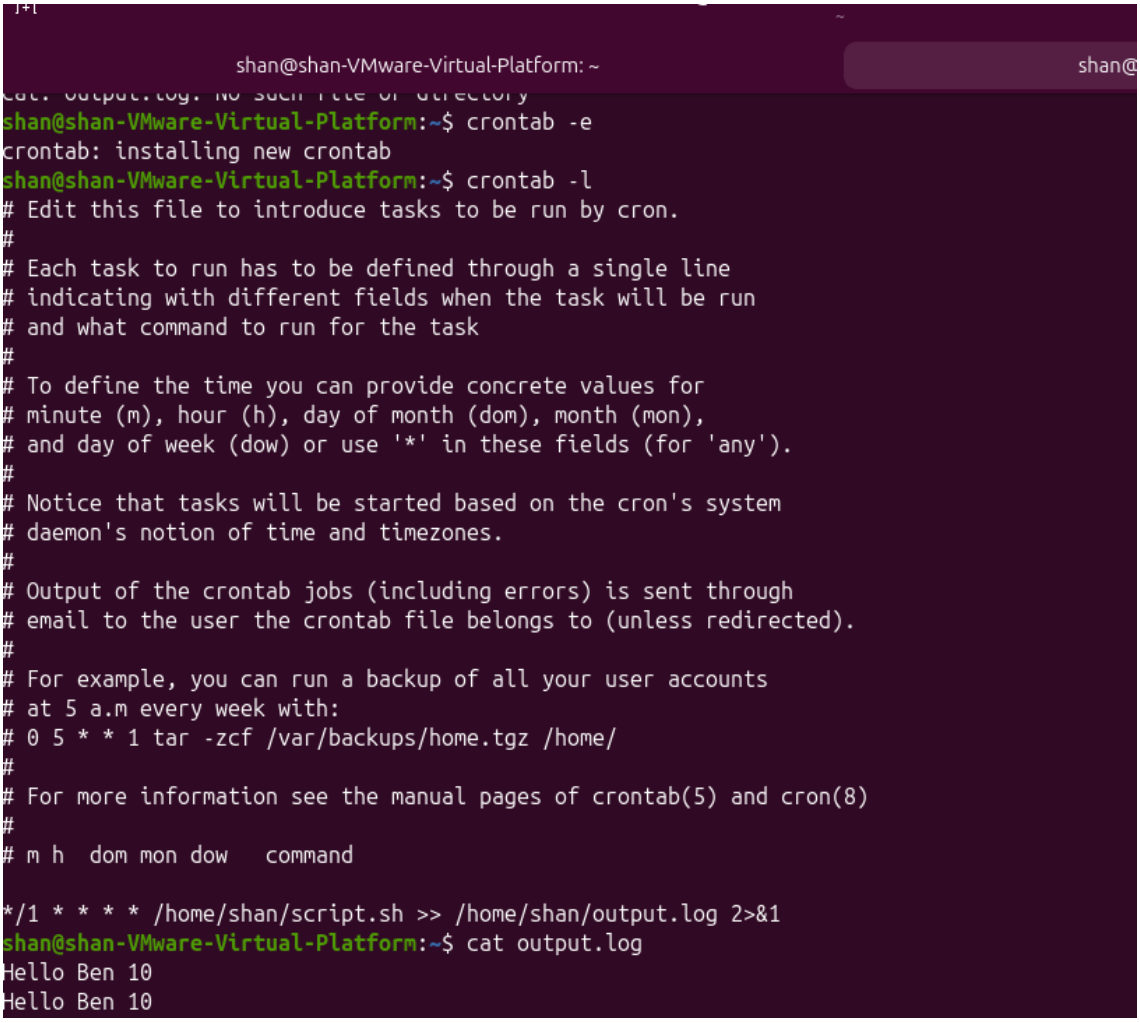
Task 1: Implement Advanced Scheduling with Prioritization

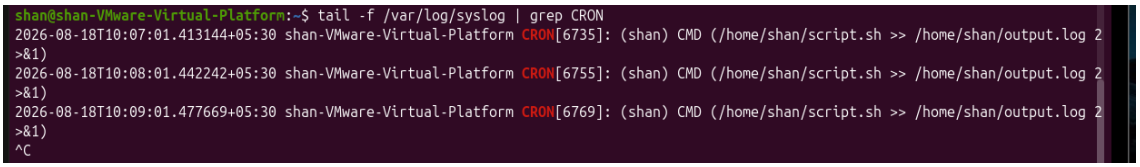
- Install cron using the command `sudo apt install cron`.
- Start the cron service using `sudo systemctl start cron`.
- Enable the cron service using `sudo systemctl enable cron`.
- Check the cron service status using `sudo systemctl status cron`.
- Go to the home directory using `cd ~`.
- Create the task script using `nano script.sh`.
- Make the script executable using `chmod +x script.sh`.
- Verify the script permissions using `ls -l script.sh`.
- Execute the script using `./script.sh`.
- Open the crontab file for task scheduling.
- Configure the task to run periodically using the cron entry `*/1 * * * * /home/shan/script.sh >> /home/shan/output.log 2>&1`.
- Redirect the scheduled task output to the log file.

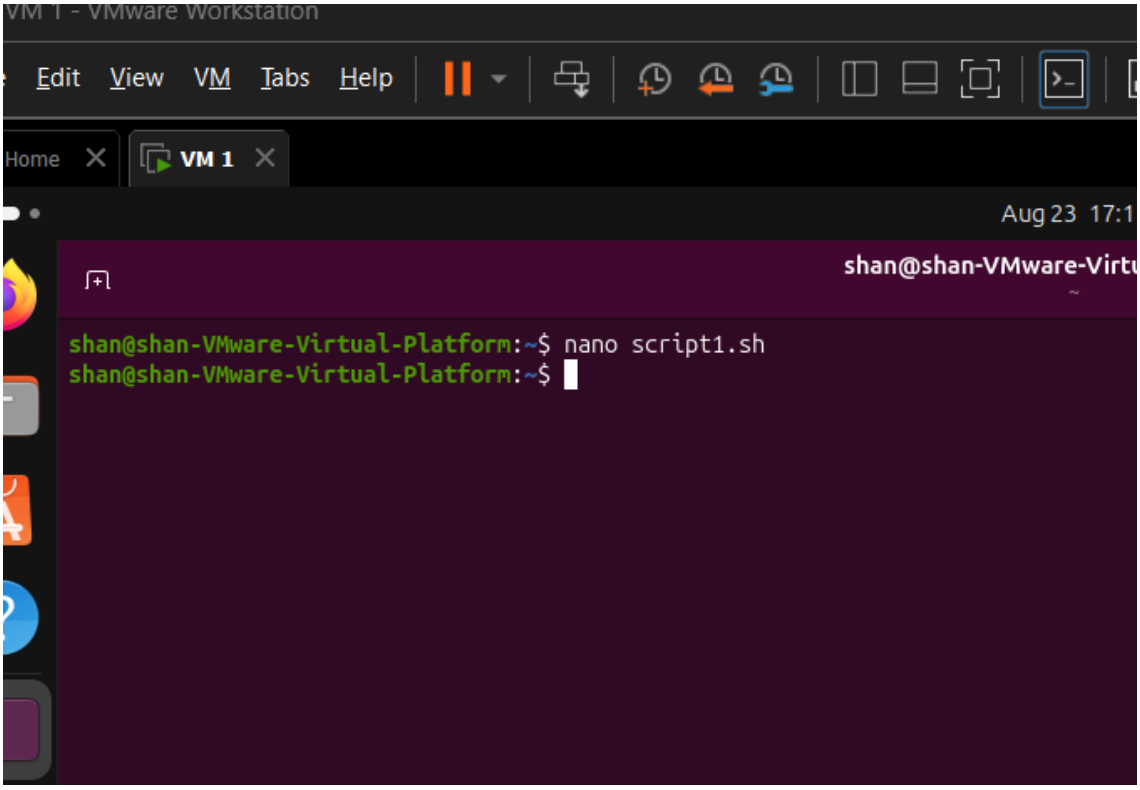
SCREENSHOTS:

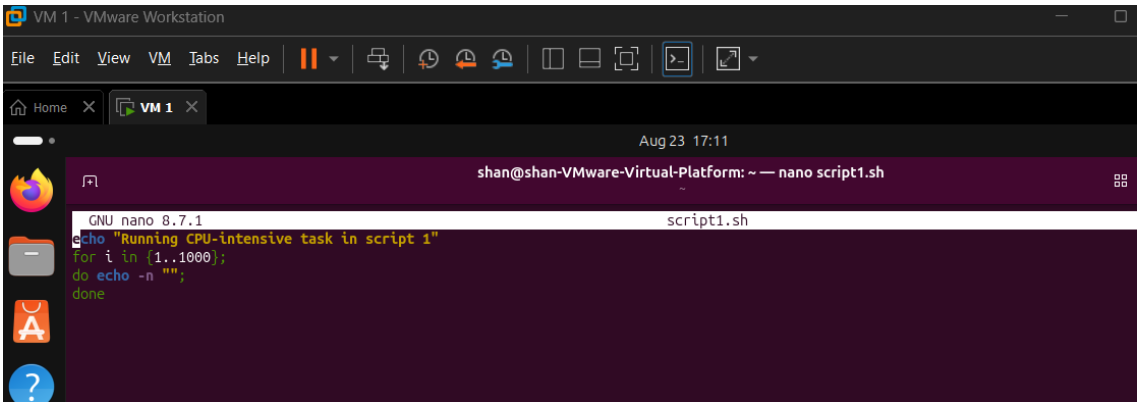


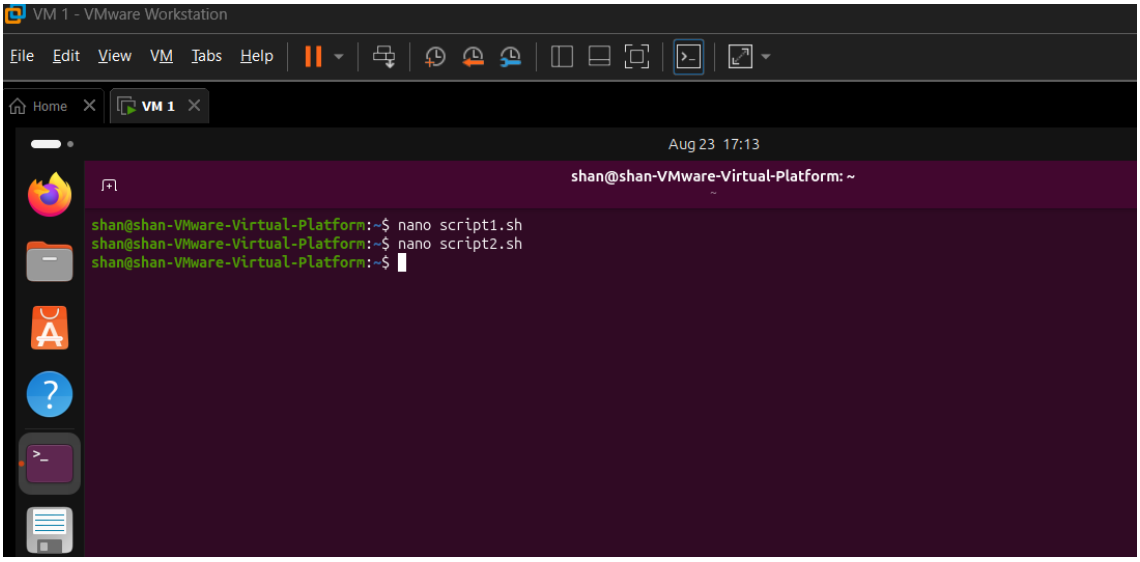


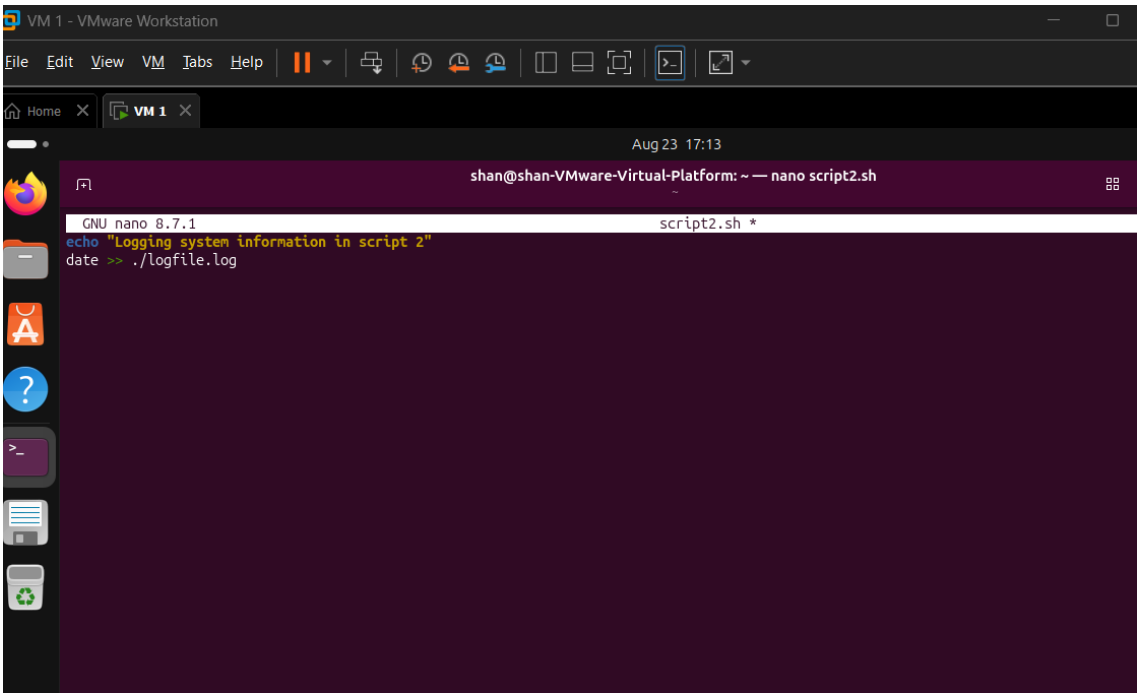

shan@shan-VMware-Virtual-Platform: ~
shan@shan-VMware-Virtual-Platform:~$ crontab -e
crontab: installing new crontab
shan@shan-VMware-Virtual-Platform:~$ crontab -l
Edit this file to introduce tasks to be run by cron.
#
Each task to run has to be defined through a single line
indicating with different fields when the task will be run
and what command to run for the task
#
To define the time you can provide concrete values for
minute (m), hour (h), day of month (dom), month (mon),
and day of week (dow) or use '*' in these fields (for 'any').
#
Notice that tasks will be started based on the cron's system
daemon's notion of time and timezones.
#
Output of the crontab jobs (including errors) is sent through
email to the user the crontab file belongs to (unless redirected).
#
For example, you can run a backup of all your user accounts
at 5 a.m every week with:
0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
For more information see the manual pages of crontab(5) and cron(8)
#
m h dom mon dow   command
*/1 * * * * /home/shan/script.sh >> /home/shan/output.log 2>&1
shan@shan-VMware-Virtual-Platform:~$ cat output.log
Hello Ben 10
Hello Ben 10


shan@shan-VMware-Virtual-Platform:~$ tail -f /var/log/syslog | grep CRON
2026-08-18T10:07:01.413144+05:30 shan-VMware-Virtual-Platform CRON[6735]: (shan) CMD (/home/shan/script.sh >> /home/shan/output.log 2>&1)
2026-08-18T10:08:01.442242+05:30 shan-VMware-Virtual-Platform CRON[6755]: (shan) CMD (/home/shan/script.sh >> /home/shan/output.log 2>&1)
2026-08-18T10:09:01.477669+05:30 shan-VMware-Virtual-Platform CRON[6769]: (shan) CMD (/home/shan/script.sh >> /home/shan/output.log 2>&1)
^C


VM 1 - VMware Workstation
Edit View VM Tabs Help
Home
VM 1
Aug 23 17:1
shan@shan-VMware-Virtu
shan@shan-VMware-Virtual-Platform:~$ nano script1.sh
shan@shan-VMware-Virtual-Platform:~$

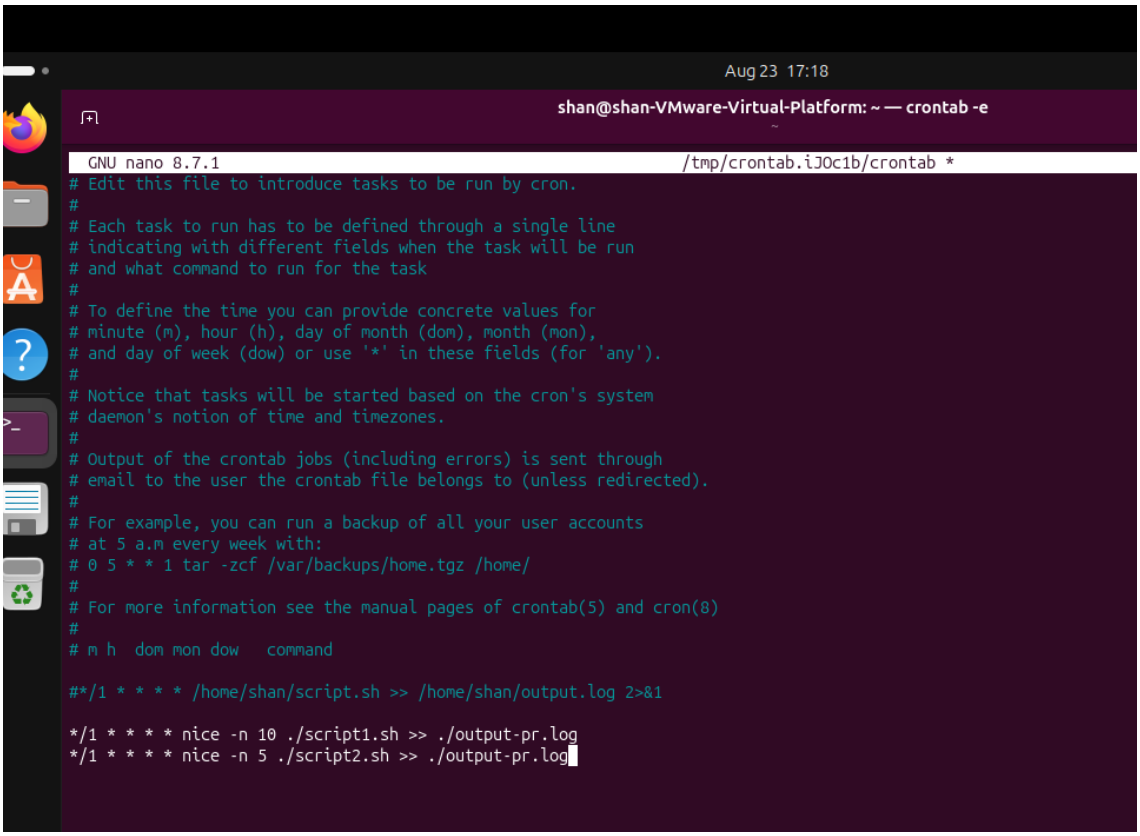

VM 1 - VMware Workstation
File Edit View VM Tabs Help
Home
VM 1
Aug 23 17:11
shan@shan-VMware-Virtual-Platform: ~ — nano script1.sh
GNU nano 8.7.1 script1.sh
echo "Running CPU-intensive task in script 1"
for i in {1..1000};
do echo -n "";
done

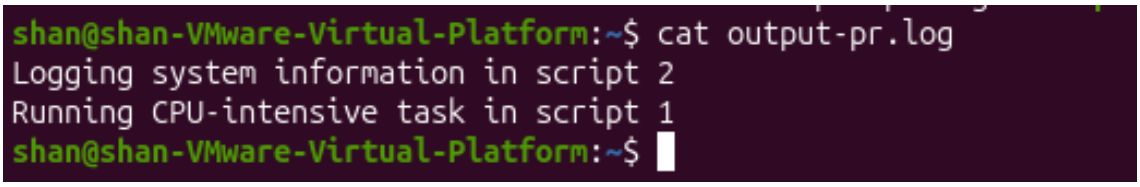

VM 1 - VMware Workstation
File Edit View VM Tabs Help
Home
VM 1
Aug 23 17:13
shan@shan-VMware-Virtual-Platform: ~
shan@shan-VMware-Virtual-Platform:~$ nano script1.sh
shan@shan-VMware-Virtual-Platform:~$ nano script2.sh
shan@shan-VMware-Virtual-Platform:~$




```
shan@shan-VMware-Virtual-Platform:~$ chmod +x script1.sh
shan@shan-VMware-Virtual-Platform:~$ chmod +x script2.sh
shan@shan-VMware-Virtual-Platform:~$
```

```
shan@shan-VMware-Virtual-Platform:~$ crontab -e
crontab: installing new crontab
shan@shan-VMware-Virtual-Platform:~$
```





OUTPUT:

CRON scheduled execution is shown in the system log.

Logging system information in script 2

Running CPU-intensive task in script 1

The scheduled command /home/shan/script.sh is executed by CRON.

RESULT:

Virtual instances with load balancing, server consolidation, and fault tolerance were successfully deployed, optimizing performance and ensuring service continuity

Exercise 7: Hypervisor Clustering in a Cloud Environment

AIM:

To understand and implement hypervisor clustering in a cloud environment , To explore high availability and load balancing in virtualized environments and To manage and migrate virtual machines within a clustered hypervisor setup.

PROCEDURE:

Part 1: Setting Up a Hypervisor Cluster

Task 1: Set Up Hypervisor Hosts

- Configure VMware Workstation Pro to use VMnet8 in NAT mode with the 192.168.94.0/24 subnet and DHCP enabled.
- Use the DHCP range 192.168.94.128–192.168.94.254.
- Create the first nested Proxmox VE virtual machine, Proxmox-01, using the AMD64 Proxmox VE ISO and connect it to the VMware NAT network.
- Install Proxmox VE with hostname pve01.cloud.local, address 192.168.94.10/24, and gateway 192.168.94.2.
- Create the second nested Proxmox VE virtual machine, Proxmox-02, with hostname pve02.cloud.local, address 192.168.94.11/24, and gateway 192.168.94.2.

Task 2: Configure Hypervisor Clustering

- Verify connectivity by pinging pve01 (192.168.94.10) from pve02 and confirm the successful test with 0% packet loss.
- Open the pve01 Proxmox web interface and go to Datacenter → Cluster.
- Create the cluster named cloud-cluster.
- Use the Join Information from pve01 to join pve02 to cloud-cluster.
- Verify that pve01 and pve02 are online nodes in the cluster.
- Verify that pve01 uses 192.168.94.10 and pve02 uses 192.168.94.11.

Part 2: Implement High Availability and Load Balancing

Task 1: Configure High Availability for VMs

- Create and install test VM 100, named ha-test, on the Proxmox cluster.
- Prepare replicated storage on both Proxmox nodes using a dedicated ZFS pool and move the VM disk to the replicated ZFS storage.
- Create a replication job for VM 100 from pve01 to pve02.
- Verify that the replication state is OK and FailCount is 0.
- Add VM 100 to Proxmox HA with the desired state started.
- Perform controlled HA migration and node-failure recovery; the supplied procedure uses a quorum adjustment as a lab workaround.

Task 2: Implement Load Balancing

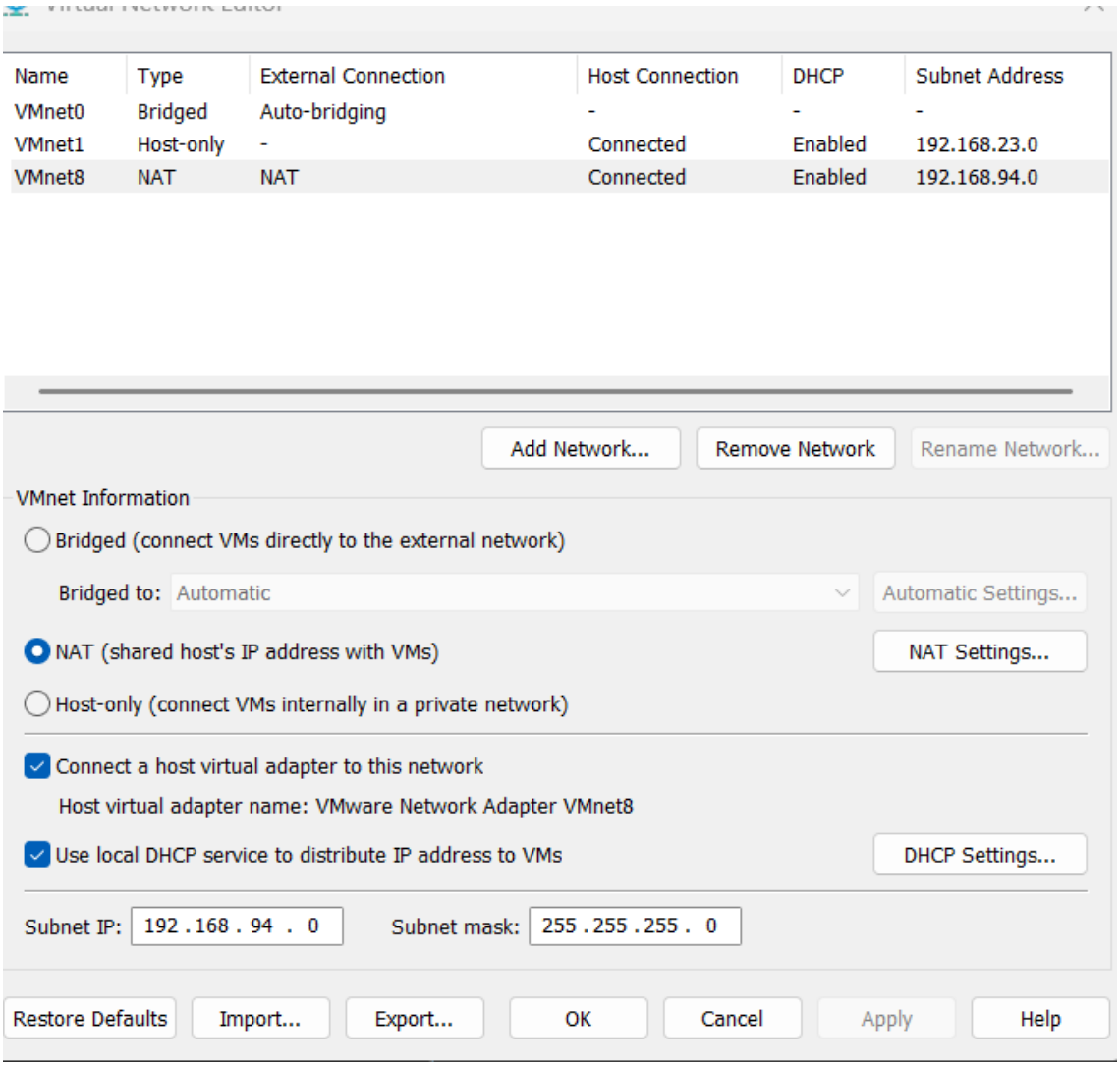
- Install HAProxy inside Debian VM 100.
- Create two lightweight HTTP backends on 127.0.0.1:8081 and 127.0.0.1:8082.
- Configure HAProxy on port 80 with roundrobin and health checks.
- Validate the HAProxy configuration and restart the service.
- Verify that the HAProxy systemd status is Active: active (running).
- Send repeated HTTP requests and verify alternating responses from Backend Server 1 and Backend Server 2.
- Stop Backend 1 and verify that Backend 2 continues to serve requests.
- Restart Backend 1 and verify that round-robin operation is restored.

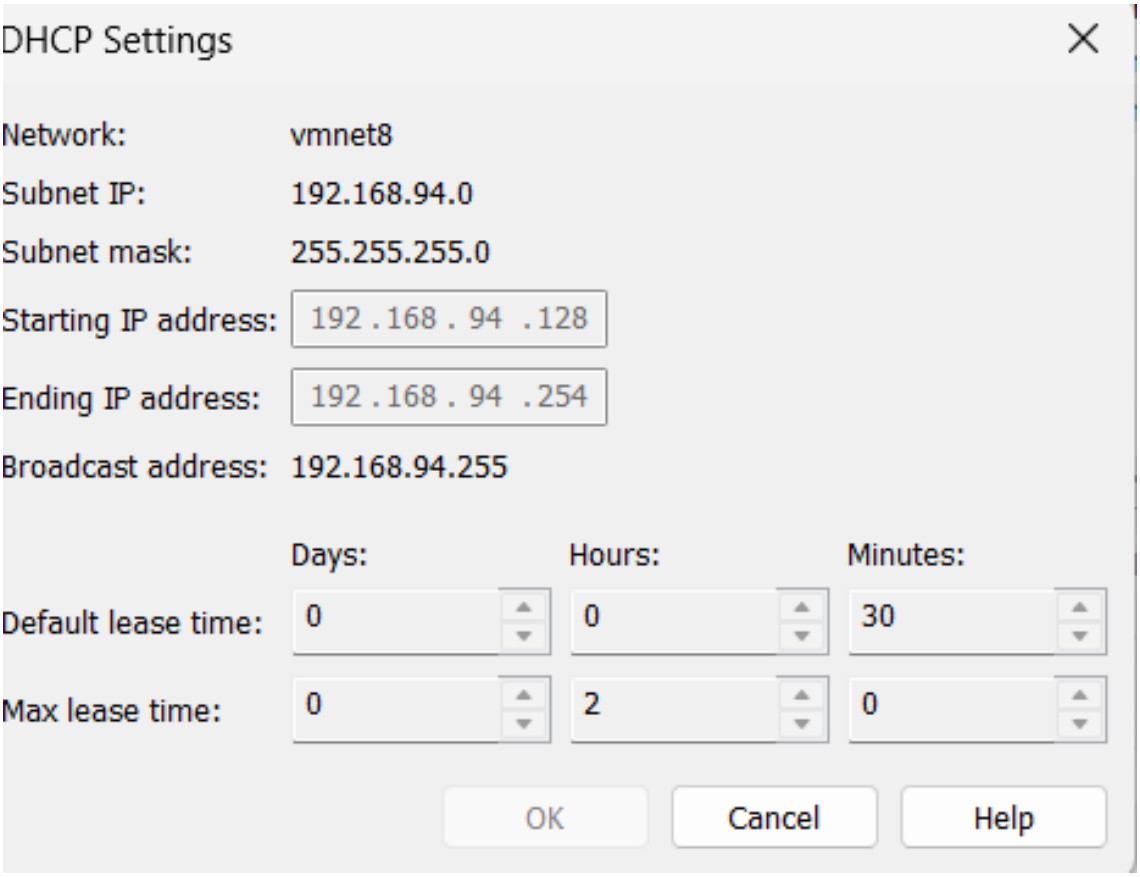
Part 3: Manage and Migrate Virtual Machines

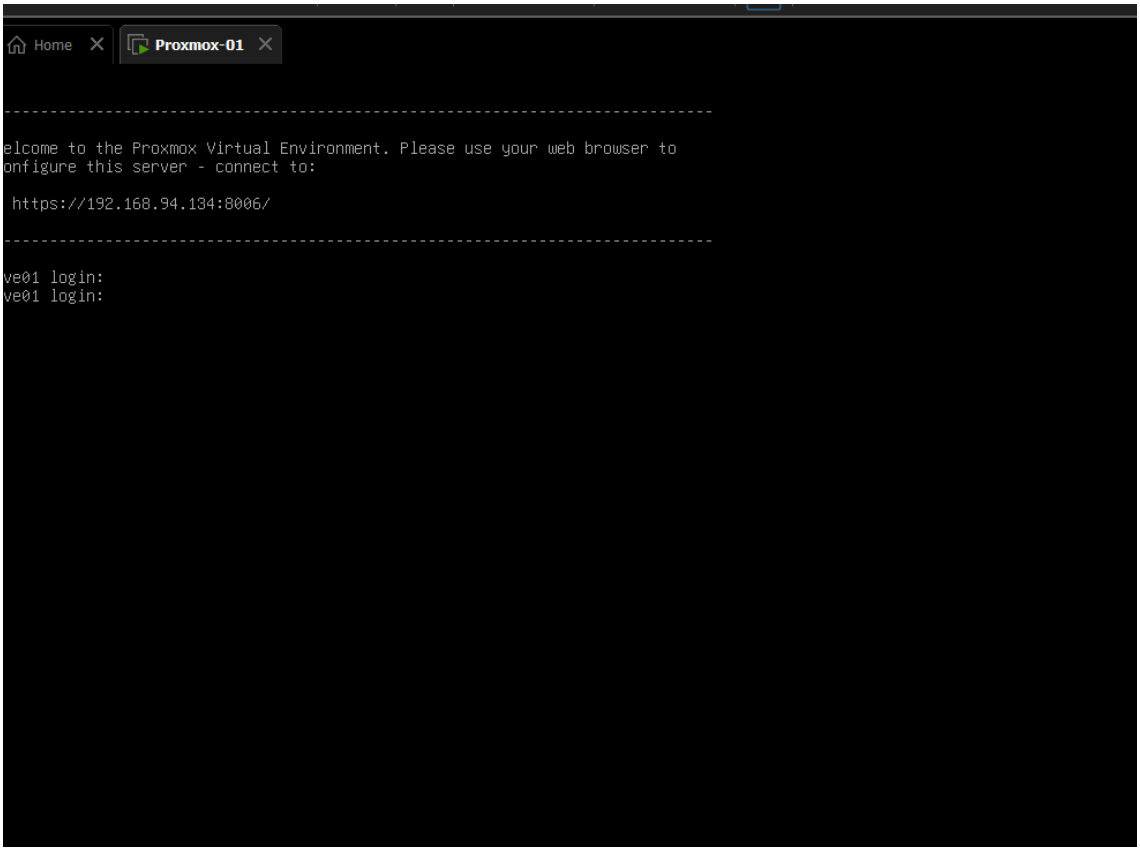
Task 1: Migrate VMs Across Hypervisors

- Log in to the Proxmox Web Interface and open Datacenter → Cluster to verify pve01 and pve02.
- Select VM 100 under pve01 and verify that it is powered on and running.
- Select VM 100 and choose More → Migrate.
- Select pve02 as the target and enable Online.
- Click Migrate and wait for the migration task to complete.
- Verify that VM 100 is associated with pve02 and remains running.

SCREENSHOTS:




DHCP Settings
Network: vmnet8
Subnet IP: 192.168.94.0
Subnet mask: 255.255.255.0
Starting IP address: 192 . 168 . 94 . 128
Ending IP address: 192 . 168 . 94 . 254
Broadcast address: 192.168.94.255
Days:
Hours:
Minutes:
Default lease time: 0 0 30
Max lease time: 0 2 0
OK
Cancel
Help


Home
Proxmox-01
elcome to the Proxmox Virtual Environment. Please use your web browser to
onfigure this server - connect to:
https://192.168.94.134:8006/
ve01 login:
ve01 login:

```
oot@pve02:~# ip addr show vmbr0
 vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group defa
   link/ether 00:0c:29:18:f1:4b brd ff:ff:ff:ff:ff:ff
   inet 192.168.94.135/24 scope global vmbr0
      valid_lft forever preferred_lft forever
   inet6 fe80::20c:29ff:fe18:f14b/64 scope link proto kernel_ll
      valid_lft forever preferred_lft forever
oot@pve02:~# ifreload -a
oot@pve02:~# ip addr show vmbr0
 vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group defa
   link/ether 00:0c:29:18:f1:4b brd ff:ff:ff:ff:ff:ff
   inet 192.168.94.11/24 scope global vmbr0
      valid_lft forever preferred_lft forever
   inet6 fe80::20c:29ff:fe18:f14b/64 scope link proto kernel_ll
      valid_lft forever preferred_lft forever
oot@pve02:~# ip route
efault via 192.168.94.2 dev vmbr0 proto kernel onlink
2.168.94.0/24 dev vmbr0 proto kernel scope link src 192.168.94.11
oot@pve02:~# ping -c 4 192.168.94.10
ING 192.168.94.10 (192.168.94.10) 56(84) bytes of data.
 bytes from 192.168.94.10: icmp_seq=1 ttl=64 time=1.02 ms
 bytes from 192.168.94.10: icmp_seq=2 ttl=64 time=0.360 ms
 bytes from 192.168.94.10: icmp_seq=3 ttl=64 time=0.383 ms
 bytes from 192.168.94.10: icmp_seq=4 ttl=64 time=0.203 ms

- 192.168.94.10 ping statistics ---
packets transmitted, 4 received, 0% packet loss, time 3055ms
t min/avg/max/mdev = 0.203/0.490/1.015/0.310 ms
```

Cluster Name: cloud-cluster Config Version: 2 Number of Nodes: 2

Cluster Nodes

odename	ID ↑	Votes	Link 0
ve01	1	1	192.168.94.10
ve02	2	1	192.168.94.11

```
service vm:100 (pve01, started)
root@pve01:~# pvesr status
JobID      Enabled    Target                       LastSync            NextSync   Duration  FailCount State
100-0      Yes        local/pve02          2026-09-12_17:45:00  2026-09-12_18:00:00   5.330695          0 OK
root@pve01:~# _
```

```
oot@ha-test:/var/www/backend2# systemctl restart haproxy
oot@ha-test:/var/www/backend2# systemctl status haproxy
 haproxy.service - HAProxy Load Balancer
     Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-09-12 17:46:47 IST; 8s ago
 Invocation: 196a67f2ecaa464aa0748940a59f8ecf
       Docs: man:haproxy(1)
             file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 5350 (haproxy)
     Status: "Ready."
      Tasks: 2 (limit: 1093)
     Memory: 55.6M (peak: 55.8M)
        CPU: 356ms
     CGroup: /system.slice/haproxy.service
             ├─5350 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
             └─5352 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock

ep 12 17:46:47 ha-test systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
ep 12 17:46:47 ha-test haproxy[5350]: [NOTICE]   (5350) : New worker (5352) forked
ep 12 17:46:47 ha-test haproxy[5350]: [NOTICE]   (5350) : Loading success.
ep 12 17:46:47 ha-test systemd[1]: Started haproxy.service - HAProxy Load Balancer.
oot@ha-test:/var/www/backend2#
```

```
Sep 12 17:46:47 ha-test systemd[1]: Started haproxy.service - HAProxy Load Balancer.
root@ha-test:/var/www/backend2# for i in {1..6}; do curl -s http://127.0.0.1; echo; done
Response from backend 1

Response from backend 2

Response from backend 1

Response from backend 2

Response from backend 1

Response from backend 2

root@ha-test:/var/www/backend2#
```

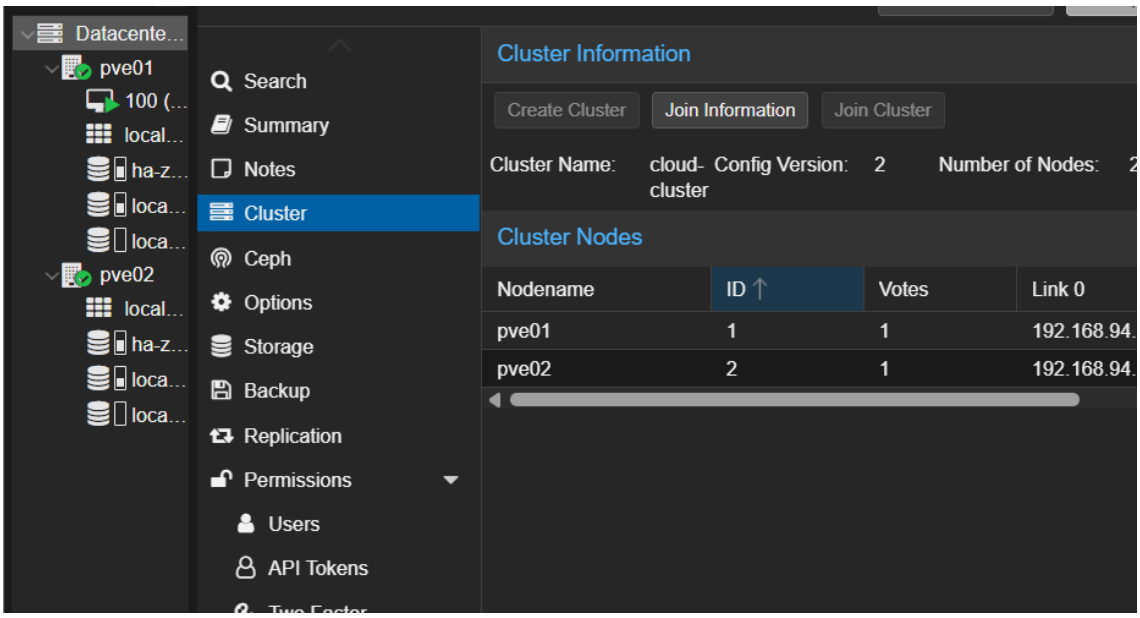
```
oot@ha-test:/var/www/backend2# hostname -I
92.168.94.137
```

```
oot@pve01:~# pvecm status
luster information
------------------
ame:             cloud-cluster
onfig Version:   2
ransport:        knet
ecure auth:      on

uorum information
-----------------
ate:             Sat Sep 12 17:53:19 2026
uorum provider:  corosync_votequorum
odes:            2
ode ID:          0x00000001
ing ID:          1.43
uorate:          Yes

otequorum information
---------------------
xpected votes:   2
ighest expected: 2
otal votes:      2
uorum:           2
lags:            Quorate

embership information
---------------------
   Nodeid      Votes Name
x00000001          1 192.168.94.10 (local)
x00000002          1 192.168.94.11
oot@pve01:~# _
```



OUTPUT:

pve01 and pve02 are online nodes in cloud-cluster.

Ping to pve01 from pve02 completed with 0% packet loss.

VM 100 replication: State = OK and FailCount = 0.

HAProxy systemd status: Active: active (running).

Requests are distributed between Backend Server 1 and Backend Server 2.

VM 100 is migrated from pve01 to pve02 and remains running.

RESULT:

Hypervisor clustering has been successfully implemented in VMWare Workstation Pro.